“INVADOPODIA , CORTACTIN AND INTRINSIC DISORDER ANALYSIS ”

GMS 7910 - DIRECTED RESEARCH

JAHNAVI MANDHADAPU

U78866713

Under the Guidance of

DR. VLADIMIR UVERSKY

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MORSANI COLLEGE OF MEDICINE

UNIVERSITY OF SOUTH FLORIDA

ABSTRACT:

Cortactin is a cytoskeletal protein crucial for regulating cell motility and cytoskeletal dynamics. Its modular structure enables interactions with actin filaments and actin-binding proteins, facilitating actin polymerization and branching. Cortactin's involvement in cellular processes like migration, adhesion, and endocytosis underscores its importance in normal cell physiology. Dysregulation of cortactin has been implicated in cancer metastasis, highlighting its potential as a therapeutic target. Post-translational modifications, such as phosphorylation, modulate cortactin activity, adding a layer of complexity to its regulatory mechanisms. Further research into cortactin's functions promises insights into cellular behavior and disease pathology.(g) In the meanwhile, however, on the opposite side Intrinsic disorder proteins (IDPs) are proteins that, in the absence of their macromolecular interactions with other proteins or RNA, lack a stable or orderly three-dimensional structure. IDPs have a higher hydrophobic properties and less net charge than regions of organised proteins. IDPs are essential for many different cellular processes and have a wide range of important roles in biological systems. While intrinsic disorder can also lead to misfolding, aggregation, and the formation of hazardous protein aggregates, it can also have positive functional benefits. Consequently, an understanding of intrinsic imperfections in specific proteins is essential. .In this research study, I invesgate the “Invadopodia, Cortactin and Intrinsic Disorder Analysis”.

In this research work, I explore the disorder regions in Cortactin using notable prediction tools: RIDAO, D2P2, FUZDROP, ALPHAFOLD and STRING. Utilising RIDAO, D2P2, FUZDROP, ALPHAFOLD and STRING, the analysis of disorder regions provides significant insights into the structural dynamics of cortactin. The discoveries enhance the understanding of the current state of disorder regions while providing an outline for future research into the underlying molecular pathways.

Keywords : Cortactin, Intrinsic disorder proteins, UNIPROT, RIDAO, D2P2, FUZDROP, ALPHAFOLD, STRING.

INTRODUCTION :

For several decades, the structural biology community focused on the single-sequence-structure-function theory as its base of operation. This theory suggests that a protein's amino acid sequence encodes for a specific native state structure, which in turn fulfils a designated function begins. Over the last 20 years, a variety of findings have disproved this idea to the point the fact that it is no longer valid. Numerous novel protein types, including intrinsic disease proteins and moonlighting proteins, have been found. Furthermore, we now know that various mescales of biomolecular dynamics in addition to structure, supervise biomolecular function, contrary to what had previously been believed to be the case. Consider that metamorphic proteins, that could fold into a wide range of distinctive structural topologies, have merely developed. (e)

INTRINSTIC DISORDER PROTEINS:

Intrinsically disordered proteins (IDPs) and regions (IDPRs) exist at the secondary or tertiary structure levels as highly dynamic ensembles of rapidly approaching conformations (or highly dynamic sets of short-lived structures), as compared with containing a three-dimensional structure. They might have

been discovered in two distinct forms: compressed (like molten globules) and extended (like coils or pre-molten globules). Ordered proteins and domains frequently perform complimentary roles that are carried out by IDPs and IDRs. Promiscuous binders referred to as IDPs and IDRs are involved in controlling and regulating of an extensive variety of cellular processes. Molecular recognition, entropic chain assemblies, molecular assembly, protein modifications, and RNA and protein chaperones are some of the broad functional categories into which these IDPs/IDRs fall. The disadvantages of it's probable that IDPs represent an issue (c). There is a correlation between IDPs and IDRs and multiple human disorders, as well as regular promiscuous interactions. It was discovered that 79% of the proteins associated with cancer were predicted to have IDRs of 30 residues or beyond (d). All disordered proteins or regions of proteins must therefore be analysed. There are several predictors available to look at the disorder areas. In this section, I'd like to talk about two predictors that I utilised in my study to identify the disorder region: Rapid Intrinsic Disorder Analysis (RIDAO)and the Database of Disordered Protein Predicted on (D2P2).

RIDAO:

RIDAO is an innovative and incredibly effective web-based predictor of disorders that enables it simpler for researchers to apply protein intrinsic disorder analysis to comparative genomics, proteomics, and genome-scale structural bioinformatics. RIDAO has the ability to analyse the entire dataset in less than an hour. At a per-residue rate predicted by the dataset. RIDAO develops a single model that virtually resembles the output of each individual of the six well-known disorder predictors.(b)

D2P2:

The D2P2 database, that provides a consensus disorder score by merging multiple cutting-edge disorder prediction algorithms, enhances the accuracy as well as reliability of the predictions. The D2P2 data's aim is to illustrate to users how well-known disorder and structure interact with one other through providing them access to the SCOP domain based on concurrent disorders. D2P2 data additionally demonstrates the inverse of a well-folded structure and may provide helpful data for improving fold forecasting techniques.(a)

FUZDROP :

A different approach computer programme and bioinformatics tool used to determine the probability of the liquid-liquid phase separation for specific protein molecules or any other molecules is known as FuzDrop. In addition, it provides the binding scores and aggregation for binding and hotspots which matches to the specific protein sequences. (f)

STRING:

STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) is a bioinformatics resource that provides information about protein-protein interactions. It integrates and annotates known and predicted protein-protein interactions from a variety of sources, including experimental data, computational predictions, and curated databases.(k) Researchers use STRING to explore functional associations between proteins, identify protein complexes, and predict protein functions based on interaction networks.(j)

ALPHAFOLD:

AlphaFold is a deep learning-based tool developed by DeepMind, a subsidiary of Alphabet Inc. It predicts the 3D structure of proteins from their amino acid sequences with remarkable accuracy.(f) This is crucial because the 3D structure of a protein largely determines its function, and accurately predicting protein structures can greatly advance our understanding of biology and accelerate drug discovery and development processes.(l)

ROLE IN BIOLOGICAL FUNCTION:

WIPF1:

Contributes to actin cytoskeleton reorganisation. plays a role in recruiting and activating WASL in conjunction with GRB2 and NCK1. Potentially involved in controlling the subcellular location of WASL, which leads to the disintegration of stress fibres and the production of filopodia. Facilitates the development of cell ruffles. Contributes significantly to the vaccinia virus's intracellular motility by acting as an adaptor to attract WASL to the virus.

Synaptopodin:

Actin-associated protein that might be involved in regulating the shape and mobility of renal podocyte foot processes and dendritic spines dependent on actin. Appears to be necessary for telencephalic neuron spines to form spine apparatuses, which are involved in synaptic plasticity.

Caldesmon:

Acts as a link between myosin and actin filaments. Actin- and myosin-binding protein involved in the control of actomyosin interactions in smooth muscle and nonmuscle cells. promotes the stabilisation of the actin filament structure by stimulating tropomyosin's actin binding. By binding to F-actin, it inhibits the activity of actomyosin ATPase in muscle tissues. Calcium-calmodulin attenuates this inhibition, whereas tropomyosin increases it. Interacts with calmodulin, actin, myosin, and two tropomyosin molecules. Performs a crucial function in both receptor capping and cellular mitosis. Participates in migration of Schwann cells in peripheral nerve regeneration

Neuromodulin:

This protein is associated with nerve growth. It is a major component of the motile 'growth cones' that form the tips of elongating axons. Plays a role in axonal and dendritic filopodia induction**.**

HEXIM 1:

General RNA polymerase II transcription inhibitory transcription regulatory transcript. An essential part of the 7SK RNP complex, P-TEFb is sequestered in a large, dormant 7SK snRNP complex together with 7SK snRNA, blocking RNA polymerase II phosphorylation and the ensuing transcriptional elongation. Potentially modulates NF-kappa-B, ESR1, NR3C1, and transcriptional activity depending on CIITA. forms the HDP-RNP complex, which provides a platform for IRF3 phosphorylation and the subsequent activation of the innate immune response via the cGAS-STING pathway, thereby contributing to the regulation of the DNA virus-mediated innate immune response.

SOX 2:

Transcription factor that, together with OCT4, forms a trimeric complex on DNA to regulate the expression of several genes, including YES1, FGF4, UTF1, and ZFP206, that are important in the development of the embryo. binds to NANOG's proximal enhancer region. Vital for the pluripotency of embryonic stem cells and for early development. SRRT target downstream that facilitates neural stem cell self-renewal. Inhibits neuronal differentiation It keeps neural cells from differentiating by blocking the action of proneural proteins. Might serve as a switch for the growth of neurons.

Cyclin Dependent Kinase:

Crucial regulator of the development of the cell cycle. has minimal inhibitory effect on CDK2 coupled to SPDYA, but inhibits the kinase activity of CDK2 attached to cyclin A. Connected to the G1 arrest. strong inhibitor of the CDK2 complexes between cyclin E and cyclin A. Interacts with cyclin type D-CDK4 complexes to form a complex and is involved in the activation, stabilisation, and assembly of the CCND1-CDK4 complex. Depending on the phosphorylation state and/or stoichiometry of the cyclin type D-CDK4 complex, functions as either an activator or an inhibitor.

AMOTL 2:

Regulates the translocation of phosphorylated SRC to peripheral cell-matrix adhesion sites. Required for proper architecture of actin filaments. Inhibits the Wnt/beta-catenin signaling pathway, probably by recruiting CTNNB1 to recycling endosomes and hence preventing its translocation to the nucleus. Participates in angiogenesis. May play a role in the polarity, proliferation and migration of endothelial cells. Selectively promotes FGF-induced MAPK activation through SRC.

HCLS 1:

Interacts with KCNC3 and the Arp2/3 complex to recruit the Arp2/3 complex to the cell cortex and control the reorganisation of the cortical actin cytoskeleton. Lessens the KCNC3 channels' rate of inactivation. Helps cells migrate through GNA13. Participates in the endocytosis pathway mediated by clathrin. possibly having an impact on the internalisation of ABC transporters like ABCB11. Might prevent CASP3 and CASP9. Increases the survival of cells. perhaps controlling calcium stores inside cells.

Actin - binding :

Effector molecules downstream that facilitate signal transmission from small GTPases and tyrosine kinase receptors to the actin cytoskeleton. contributes to the control of cytoskeletal organisation and cell shape. Essential for regulating the morphology of cells.

Right now, A detailed comprehension of the disorder regions present in Cortactin mainly focusing in Humans is necessary to comprehend the molecular mechanisms underpinning their unique structural and functional characteristics, as well as their functional diversity and dynamic behaviour. Thus, in this study, I used the predictor tools RIDAO, D2P2, FUZDROP, and ALPHAFOLD and, STRING to analyse the disorder regions in Cortactin in Humans. Specific Cortactin protein disorder regions are provided by these predictor tools.

MATERIALS AND METHODS:

- Conducted a search of comprehensive literature review articles and research articles on Cortactin.
- Found that there were many genes in different species like Human, Mouse, Rat, Pig and so on.
- Conducted a search of comprehensive literature review articles and research articles mainly on Human genes.
- Gathered information about Human Coratctin gene (CTTN) and retrived the FASTA sequences from UNIPORT.
- This FASTA sequence is further used for disorder region analysis by usin RIDAO and STRING.

Analysis of Proteins Disorder Region:

- FATSA sequence is used for the analysis of disorder region.
- Use of RIDAO, D2P2, FuzDrop, Alphafold and String predictors tools for the analysis of disorder profile.
- In String predictor tool, took 500 interactions of CTTN gene in Human and downloaded the multi-sequence FASTA format file.
- This file is further used for next steps like D2P2, FuzDrop, Alphafold and String predictors tools, I searched out the disordered region in 10 selected different proteins.

RESULTS:

Protein Disorder Region Analysis:

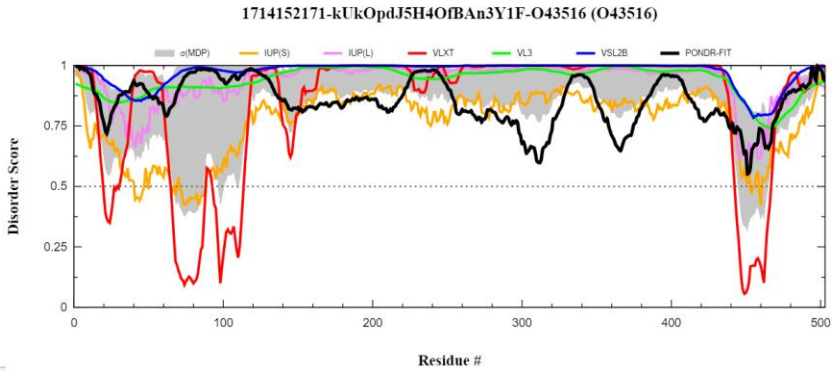
I selected 5 individual predictors named RIDAO, D2P2,FuzDrop, Alphafold and String for the analysis of the Protein Disorder region in 10 different types Cortactin Proteins in humans. Using RIDAO, I was able to find out disorder region for all the proteins in every type. By using D2P2, I got outcomes for protein sequences . By using FuzDrop, I got outcomes for 10 protein sequences for liquid-liquid phase separation. By using Alohafold, I got outcomes of 10 protein 3D structures for coratctin gene interactions in humans. And, by using the string, I got the outcomes of 10 different interactions of each protein of the cortactin gene in human.

RIDAO Predictor tool:

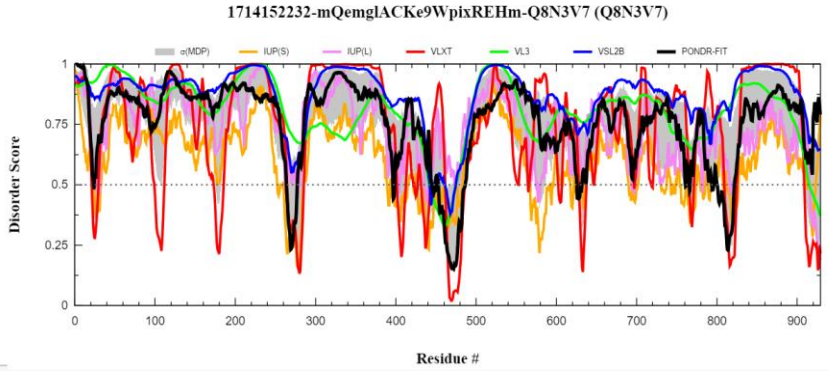
RIDAO is a recently developed, very accurate web-based disease region predictor. It provides a single combined complete graph for disorder region, indica ng disorder profile, and contains the predictors VLXT, VSLB2B, VL3, IUP_S, IUP_L, PFIT, and MDP. Addi onally, it provides the percentage of fungi proteins that are disordered by u lising various predictors. Percentage of abnormality in proteins according to various factors.

Following are the Results for Proteins using RIDAO predictor tool:

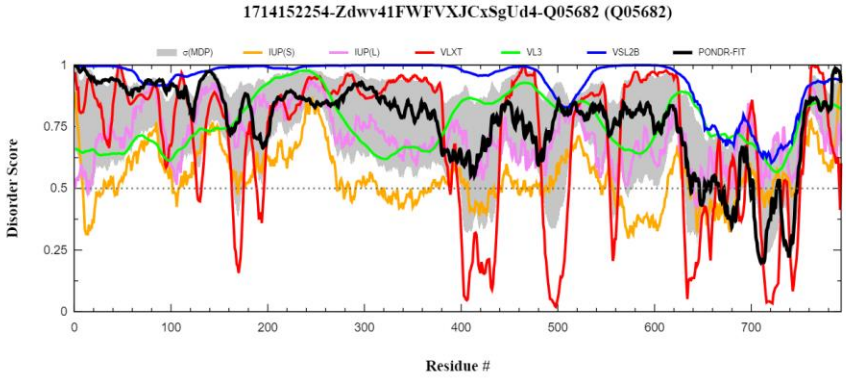
WIPF1



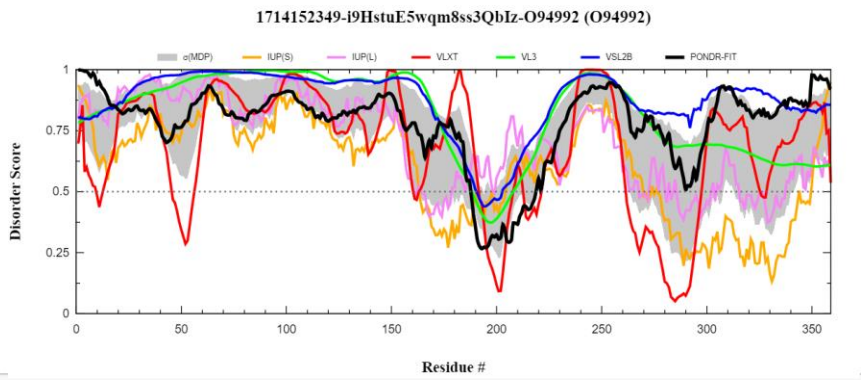
Synaptopodin



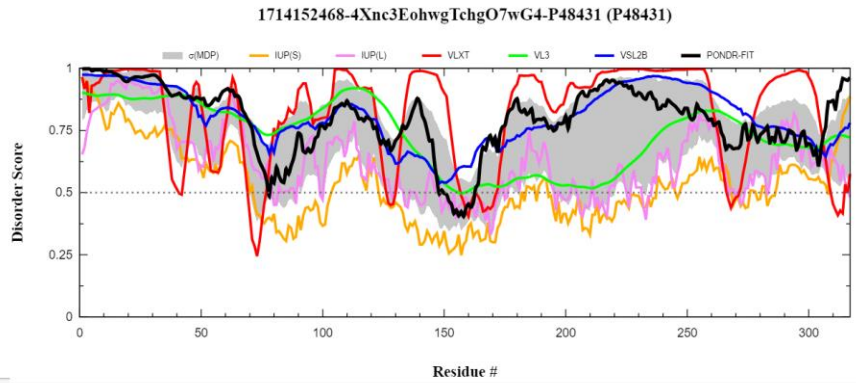
CALD1



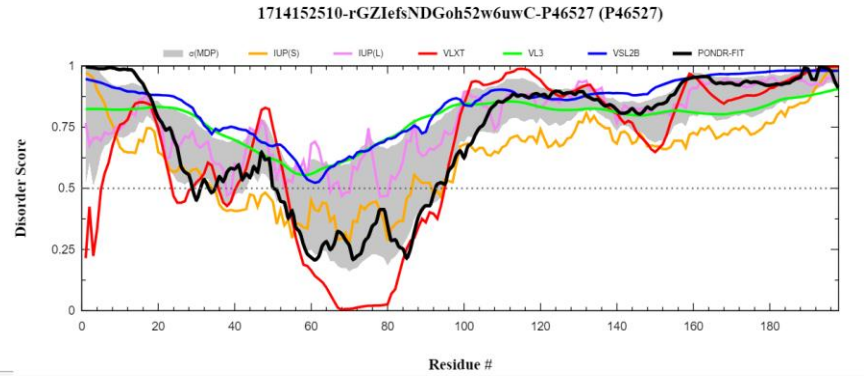
HEXIM1



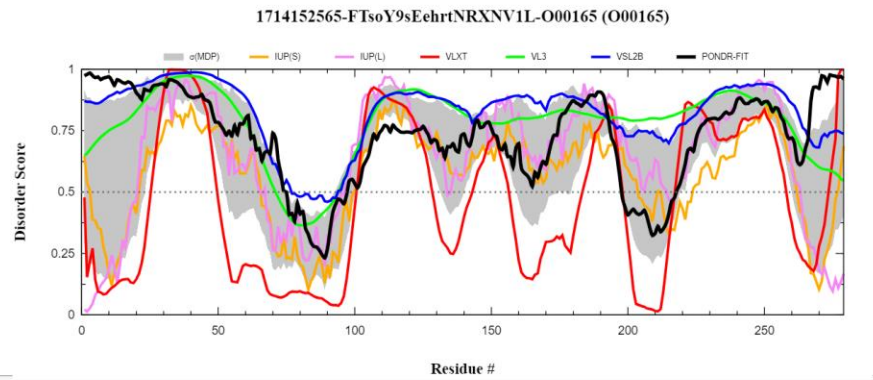
SOX2



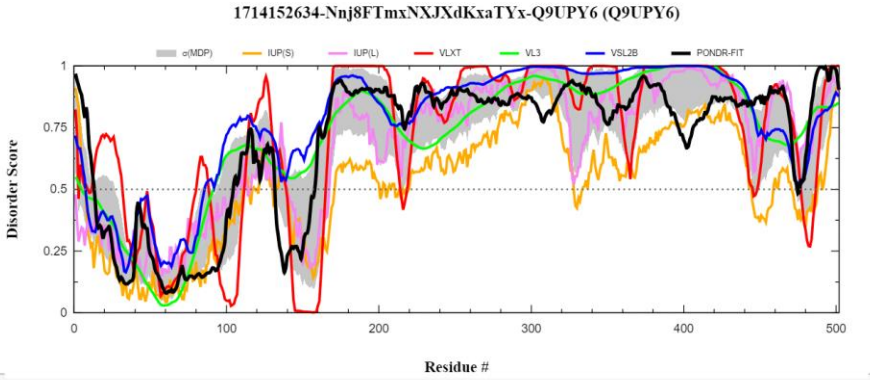
Cyclin-dependent



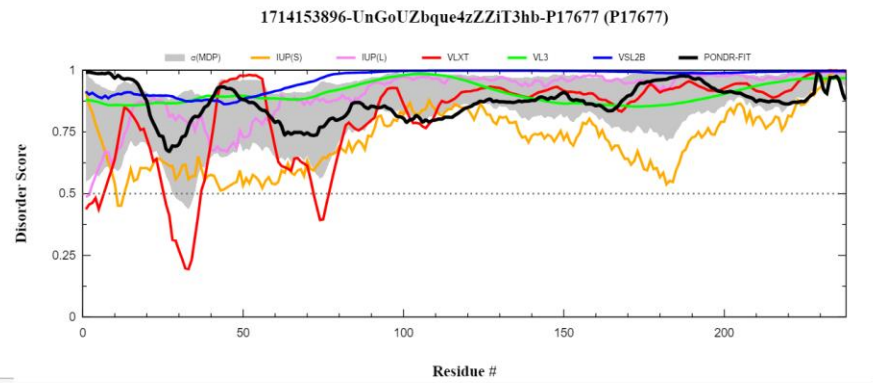
HCLS1

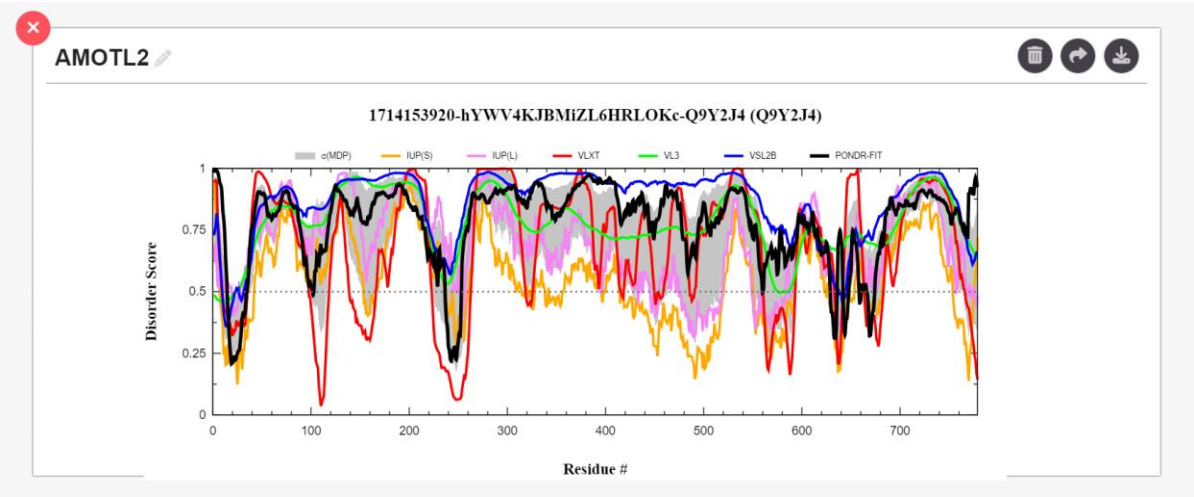


Actin-binding



GAP43



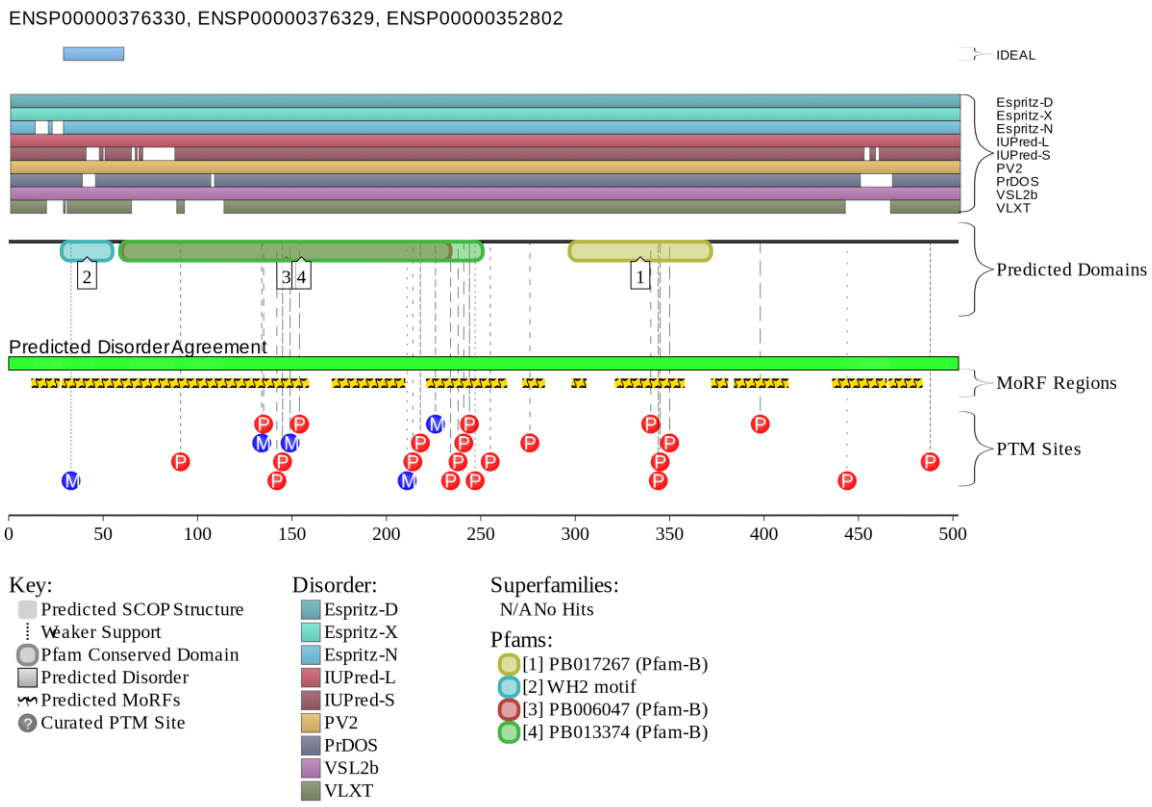


D2P2 Predictor tool:

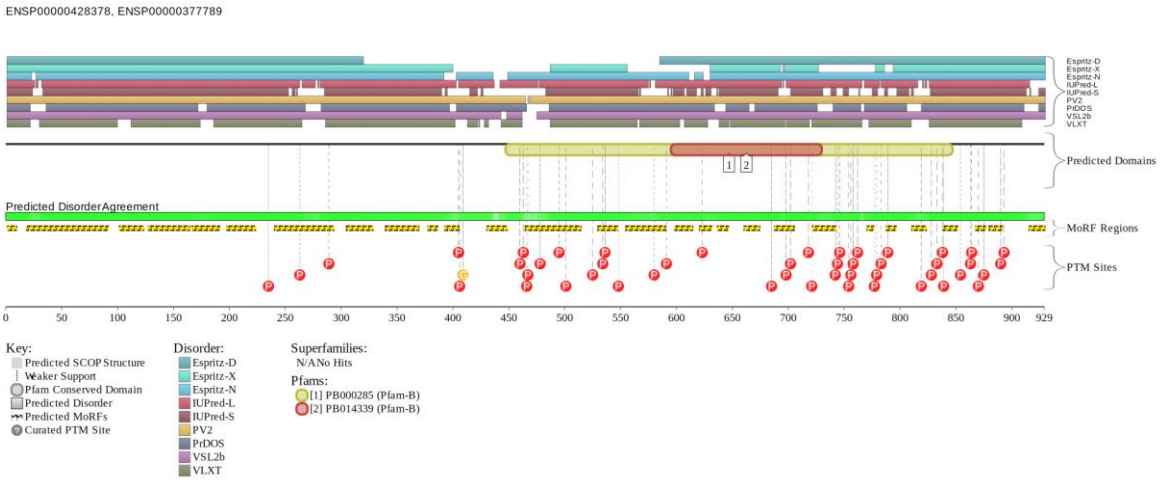
The D2P2 database combines a number of cu ng-edge disorder prediction algorithms to provide a consensus disorder score. It includes the predictors VLXT, VSL2B, PrDOS, PV2, IUPred and Espritz. Moreover, it also predicts SCOP domain from SUPERFAMILY. It gives results which show the output of various disorder predictors, posi on of func onal domains, MoRFs and PTMs.

Following are the Results for Proteins using D2P2 predictor tool:

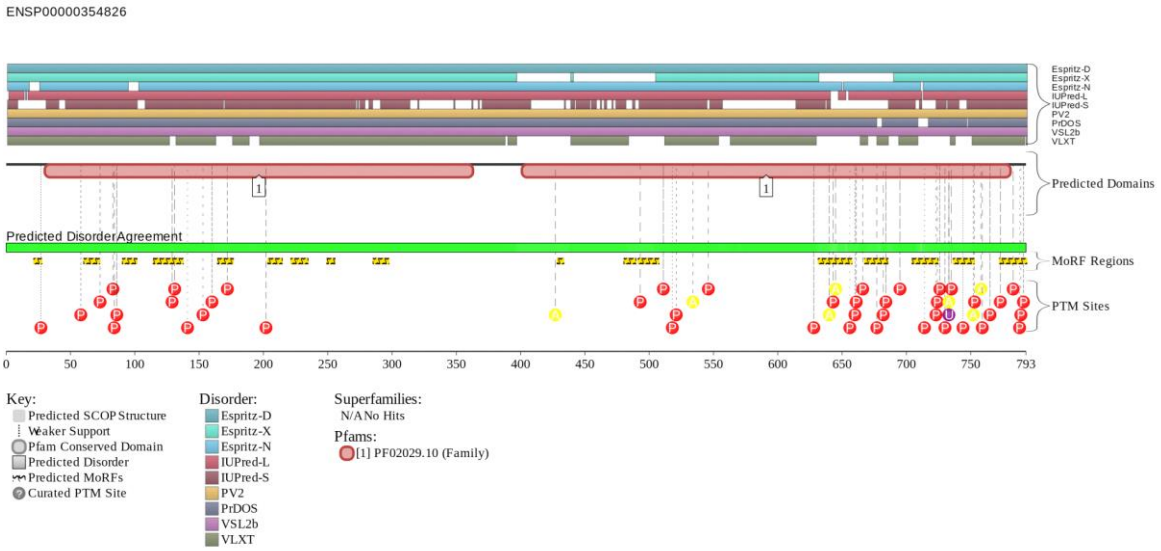
WIPF 1:



Synaptopodin:

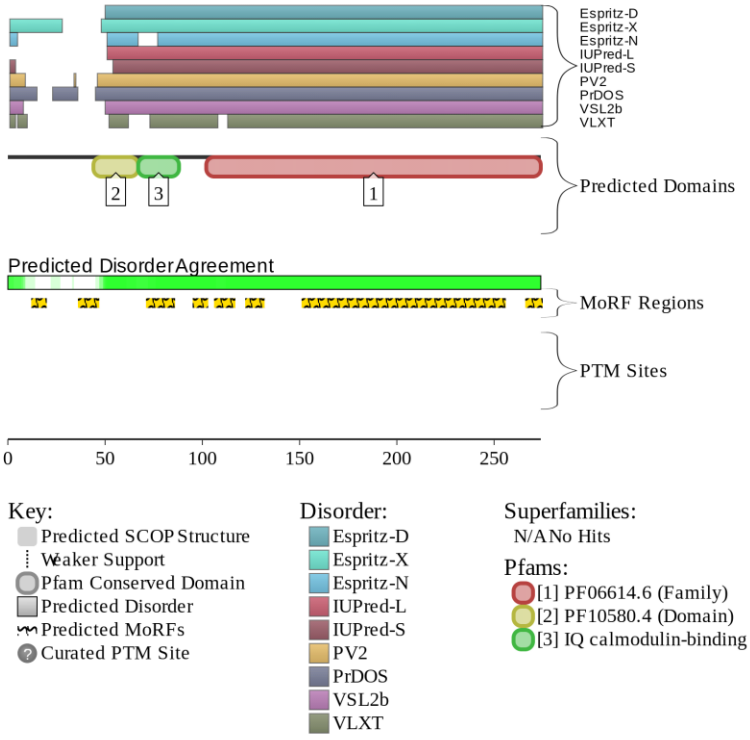


Caldesmon:



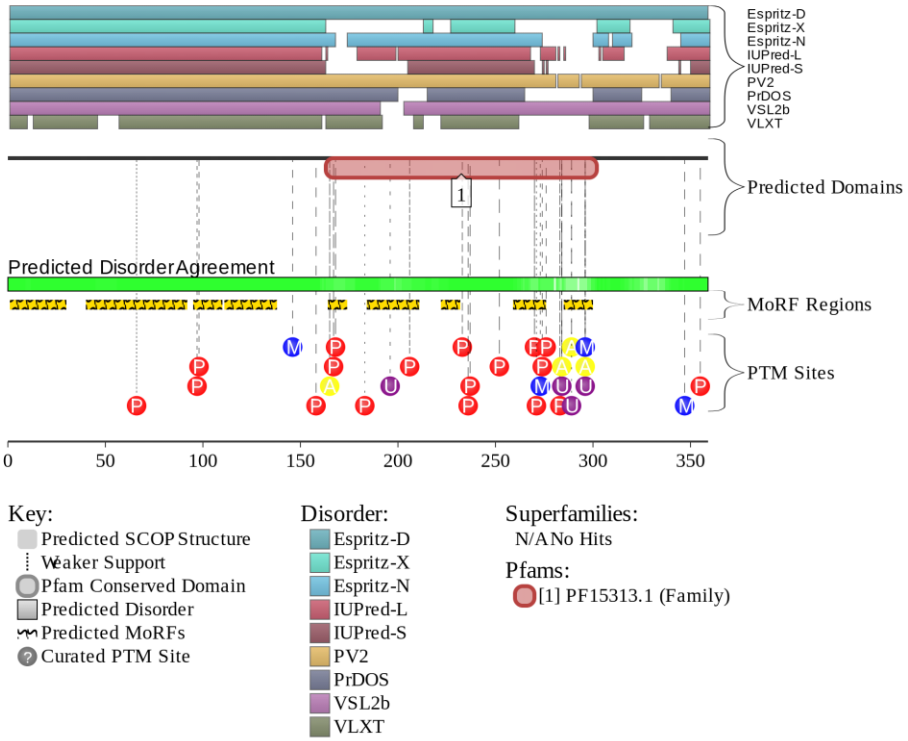
Neuromodulin:

ENSP00000377372



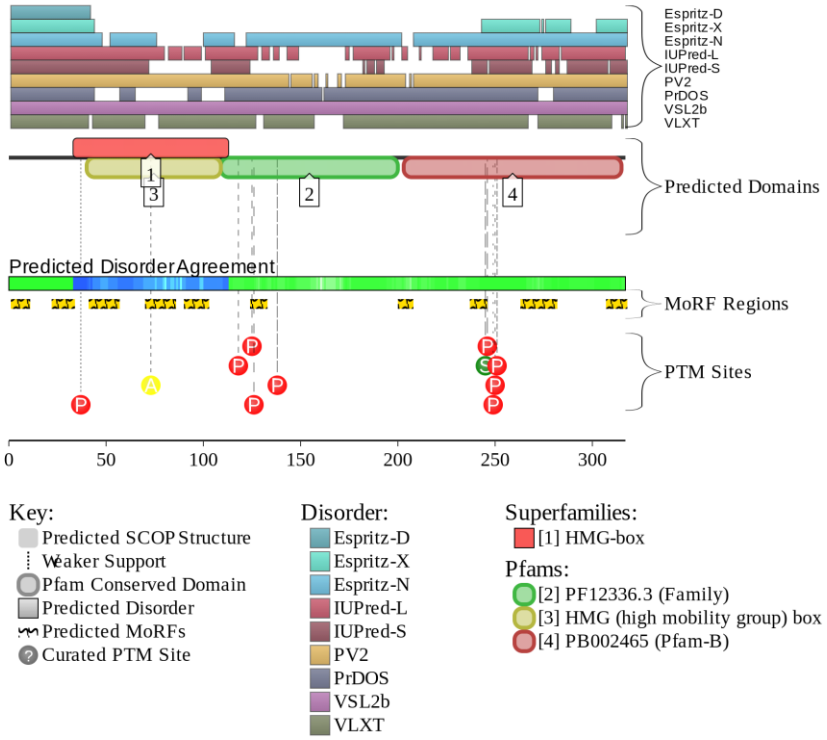
HEXIM 1:

ENSP00000328773



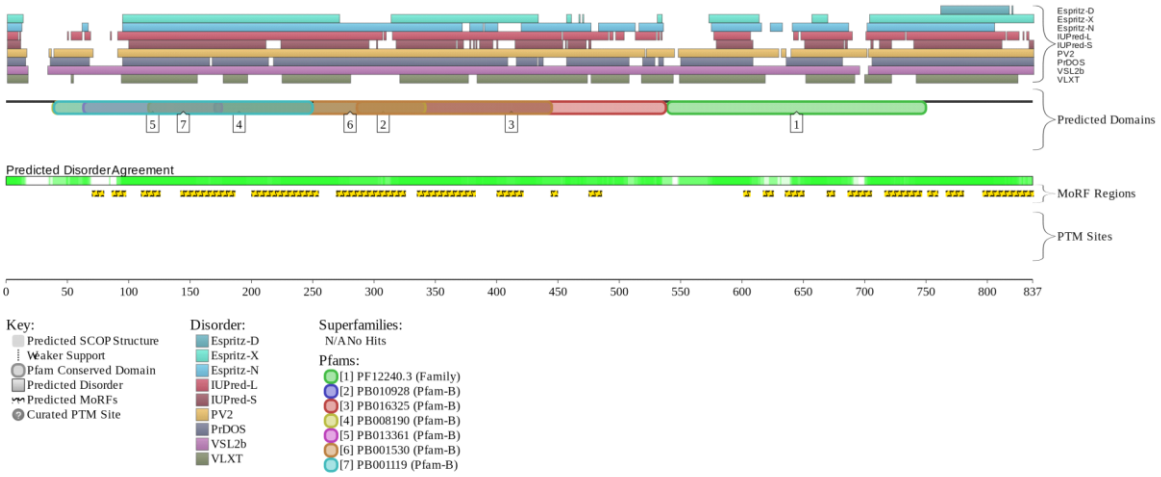
SOX 2:

ENSP00000323588



AMOTL 2:

ENSP00000424765

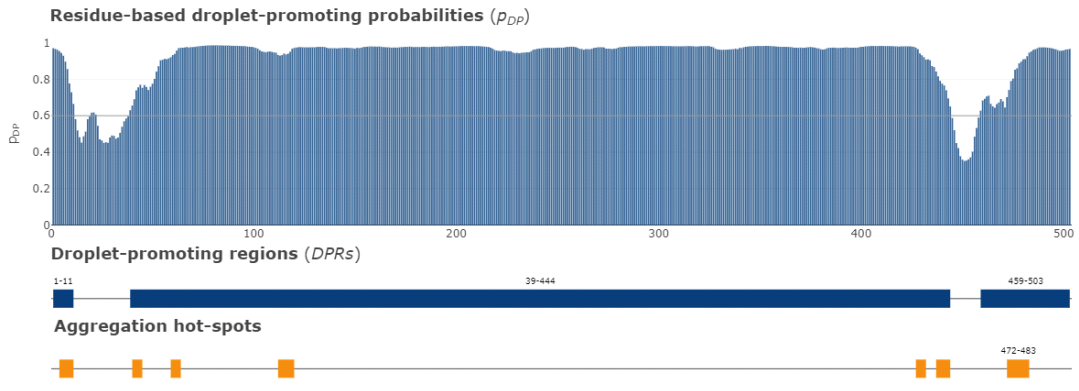


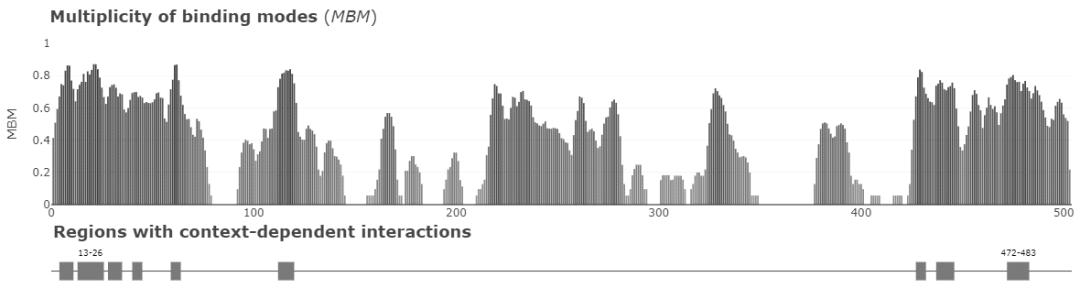
FuzDrop Predictor tool:

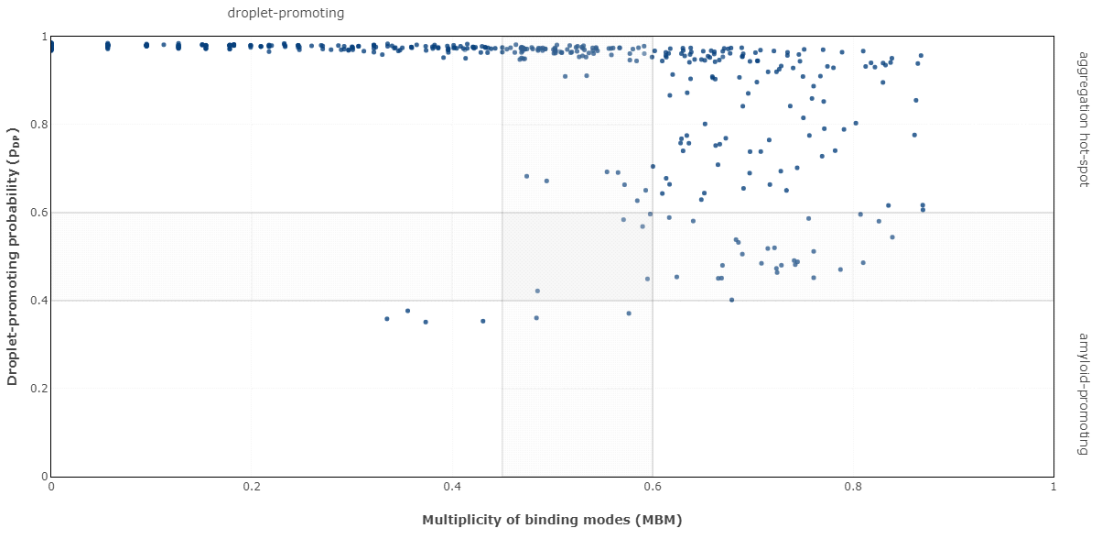
FuzDrop is a method to check the probability of proteins for liquid-liquid phase separa on and sequence-based iden fica on of both droplet-promo ng and of aggrega on-promo ng regions within droplets.It also includes the predictors like Mul plicity of Binding modes and Droplet landscape and binding scores.

Following are the Results for Proteins using FUZDROP predictor tool:

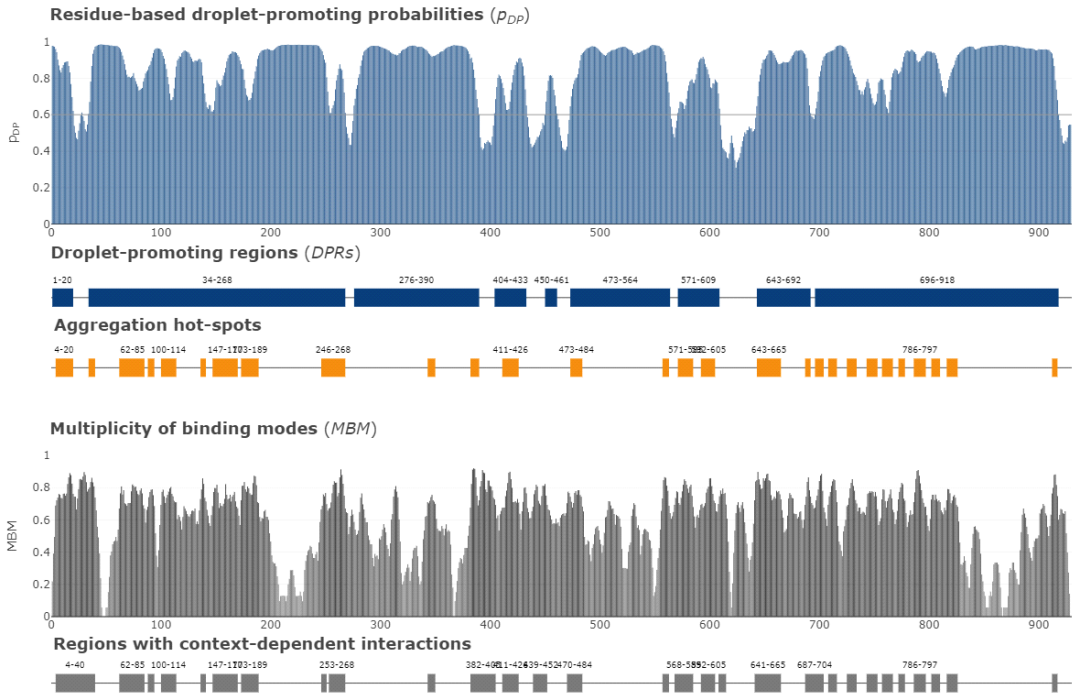
WIPF 1:

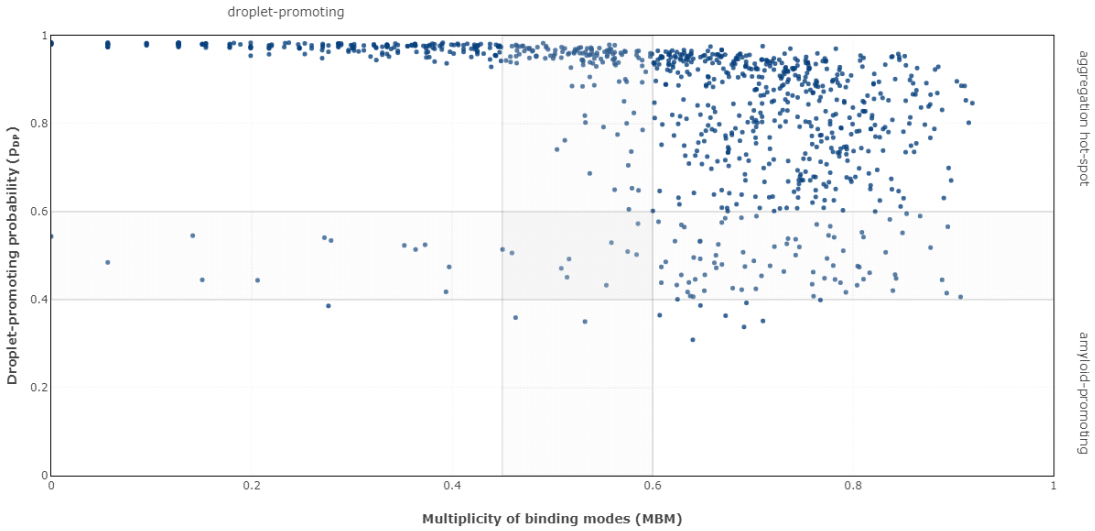




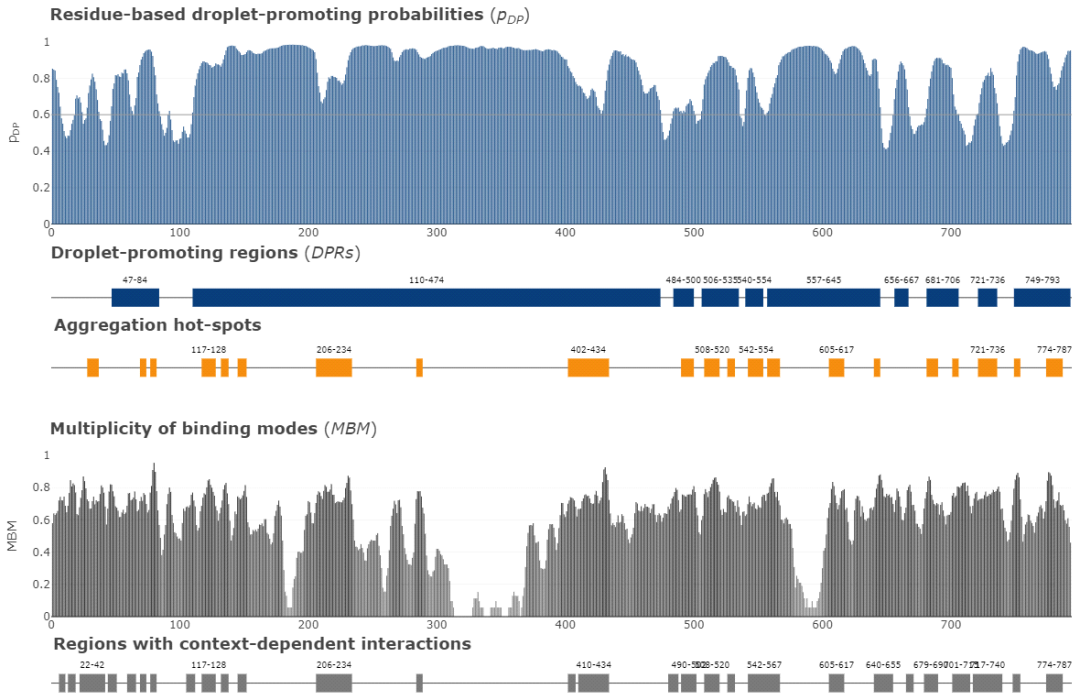


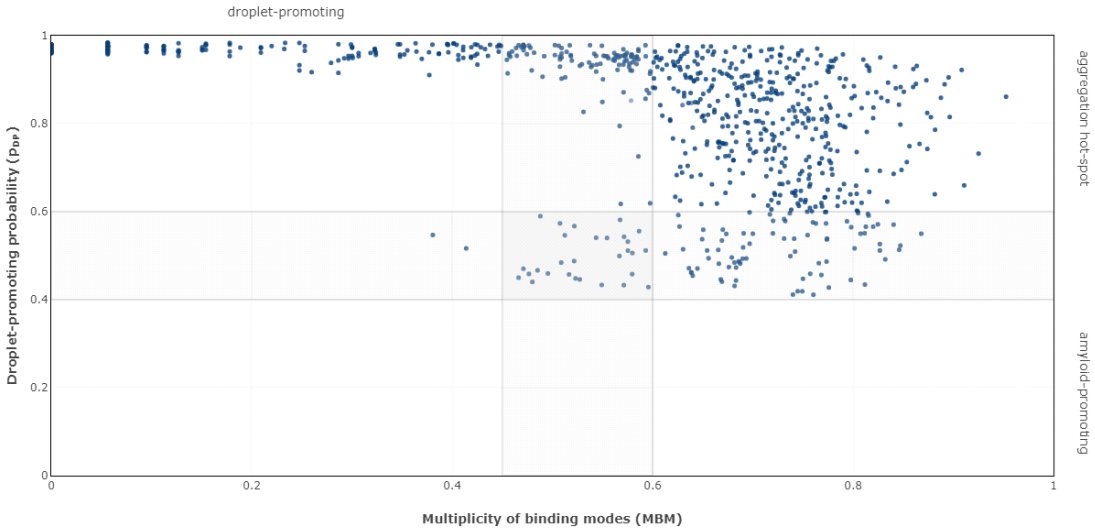
Synaptopodin:



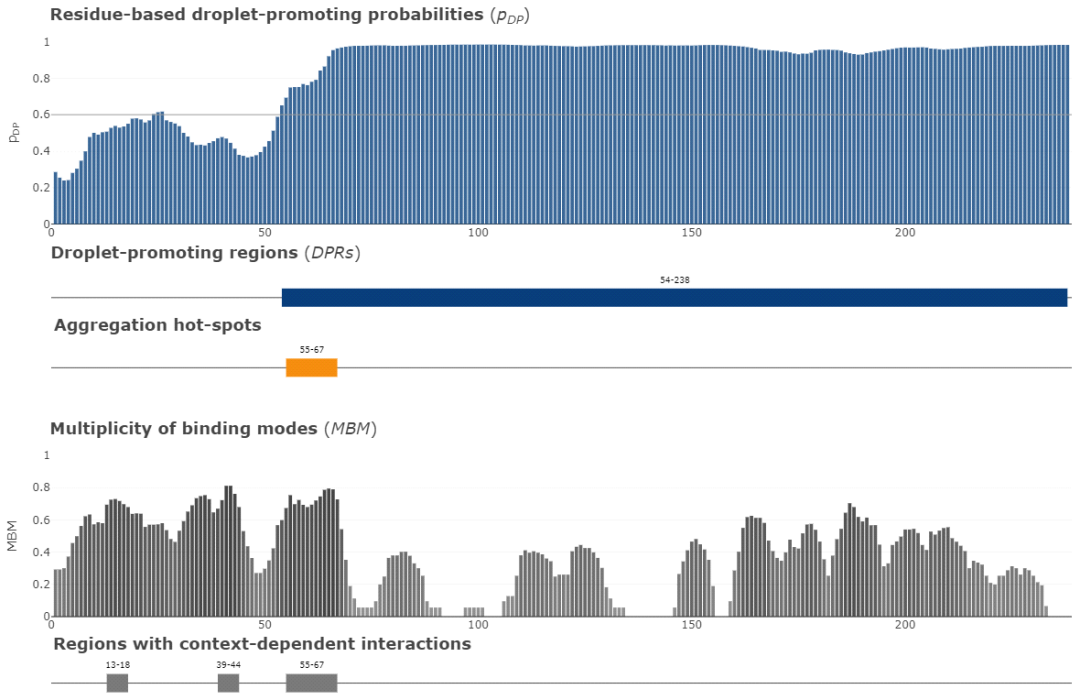


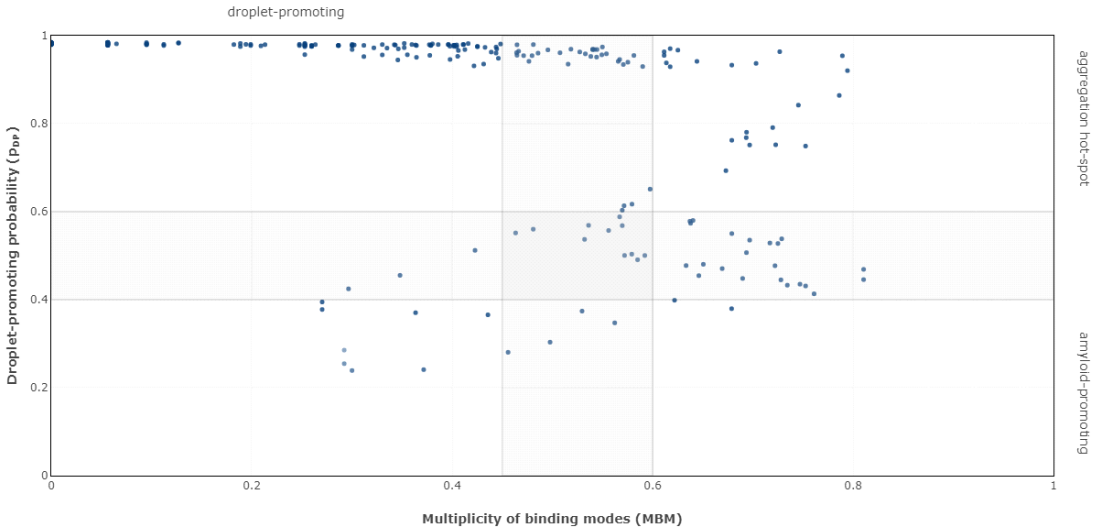
Caldesmon:


Residue-based droplet-promoting probabilities (p_{DP})
p_{DP}
Droplet-promoting regions (*DPRs*)
47-84
110-474
484-500
506-535
540-554
557-645
656-667
681-706
721-736
749-793
Aggregation hot-spots
117-128
206-234
402-434
508-520
542-554
605-617
721-736
774-787
Multiplicity of binding modes (*MBM*)
MBM
Regions with context-dependent interactions
22-42
117-128
206-234
410-434
542-567
605-617
640-655
774-787


droplet-promoting
Droplet-promoting probability (p_{DP})
Multiplicity of binding modes (MBM)
aggregation hot-spot
amyloid-promoting

Neuromodulin:

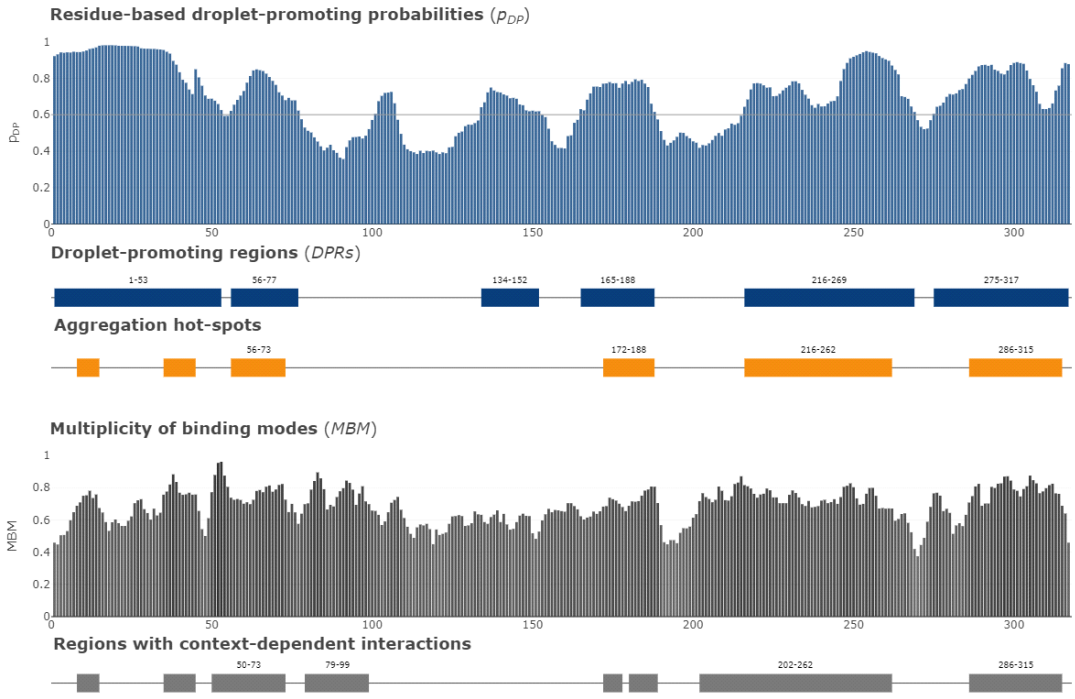


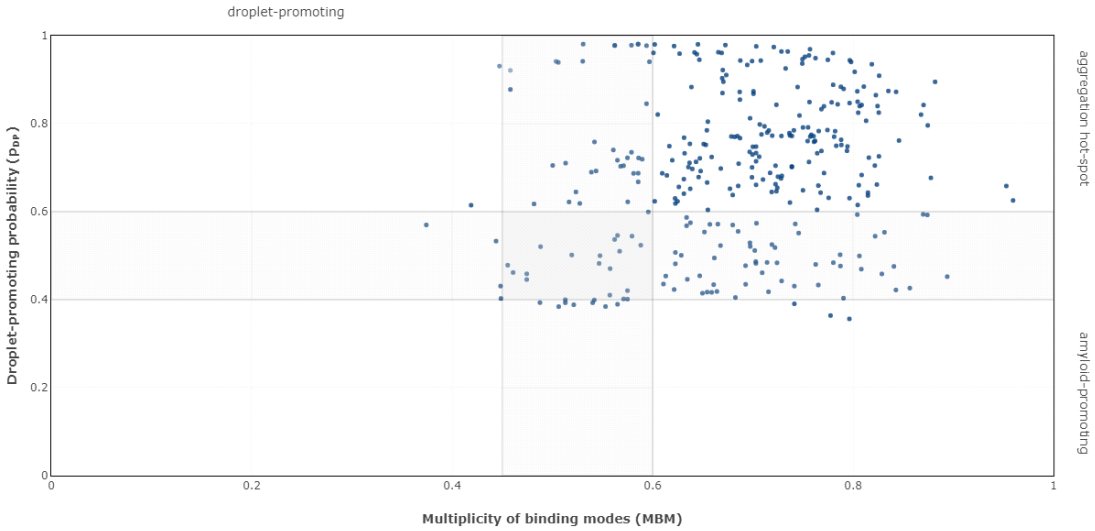


HEXIM 1:

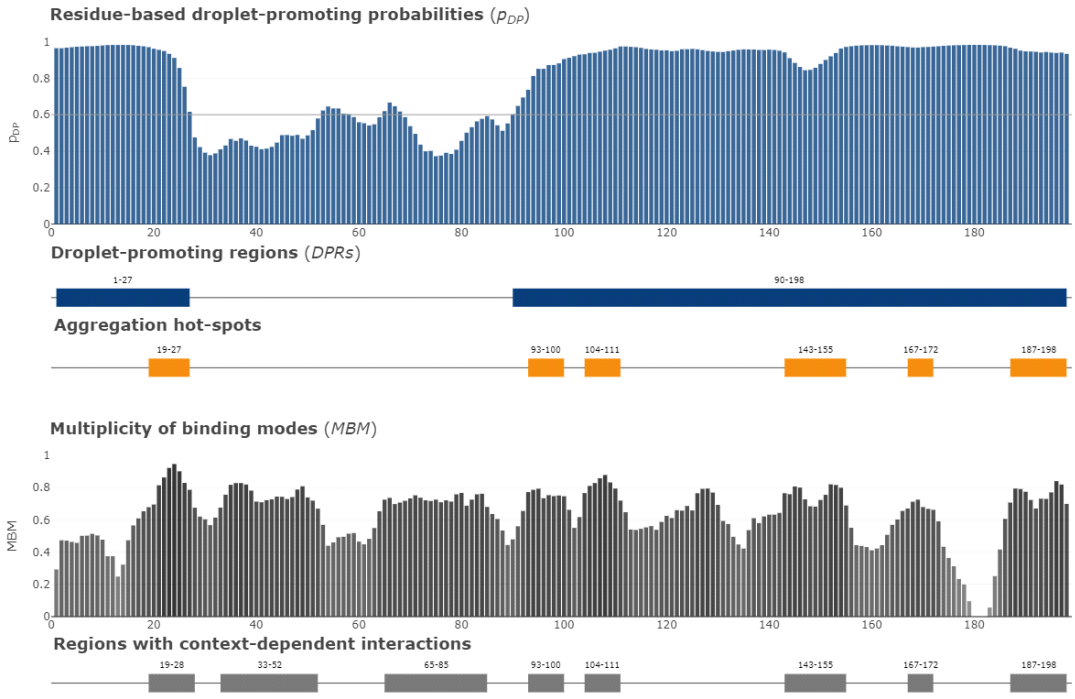
Residue-based droplet-promoting probabilities (p_{DP})
p_{DP}
Droplet-promoting regions (DPRs)
1-14
32-169
190-270
339-359
Aggregation hot-spots
32-52
70-87
99-110
145-158
191-203
215-233
Multiplicity of binding modes (MBM)
MBM
Regions with context-dependent interactions
262-285
301-312
droplet-promoting
aggregation hot-spot
amyloid-promoting
Droplet-promoting probability (p_{DP})
Multiplicity of binding modes (MBM)

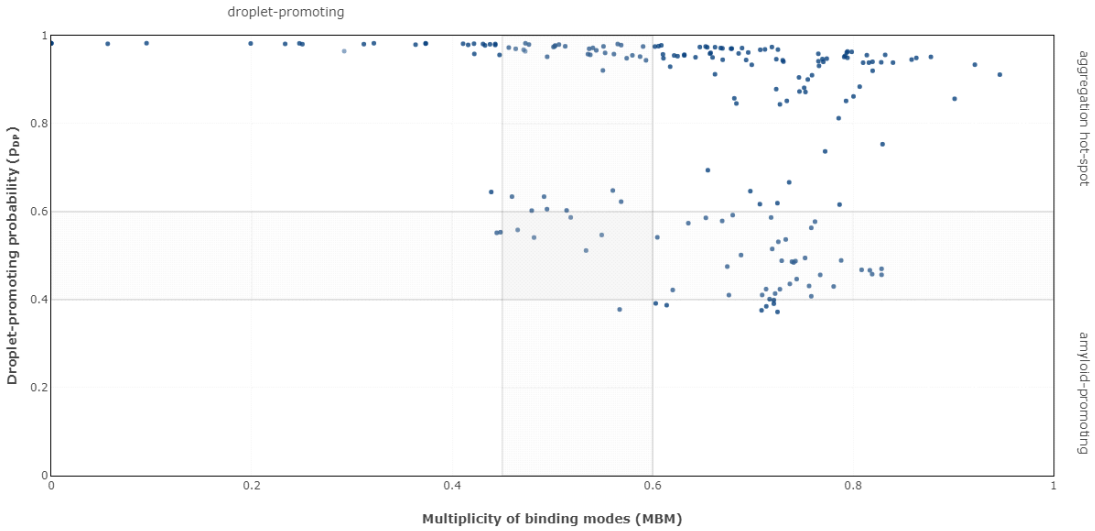
SOX 2:


Residue-based droplet-promoting probabilities (p_{DP})
p_{DP}
Droplet-promoting regions (DPRs)
1-53
56-77
134-152
165-188
216-269
275-317
Aggregation hot-spots
56-73
172-188
216-262
286-315
Multiplicity of binding modes (MBM)
MBM
Regions with context-dependent interactions
50-73
79-99
202-262
286-315

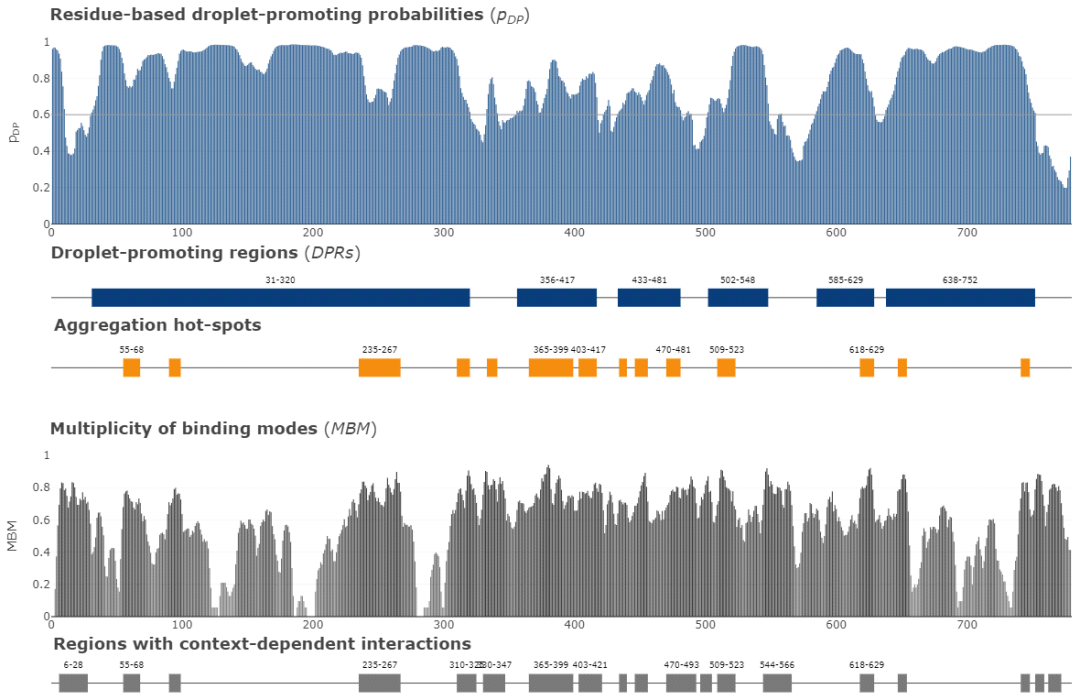

droplet-promoting
aggregation hot-spot
amyloid-promoting
Droplet-promoting probability (p_{DP})
Multiplicity of binding modes (MBM)

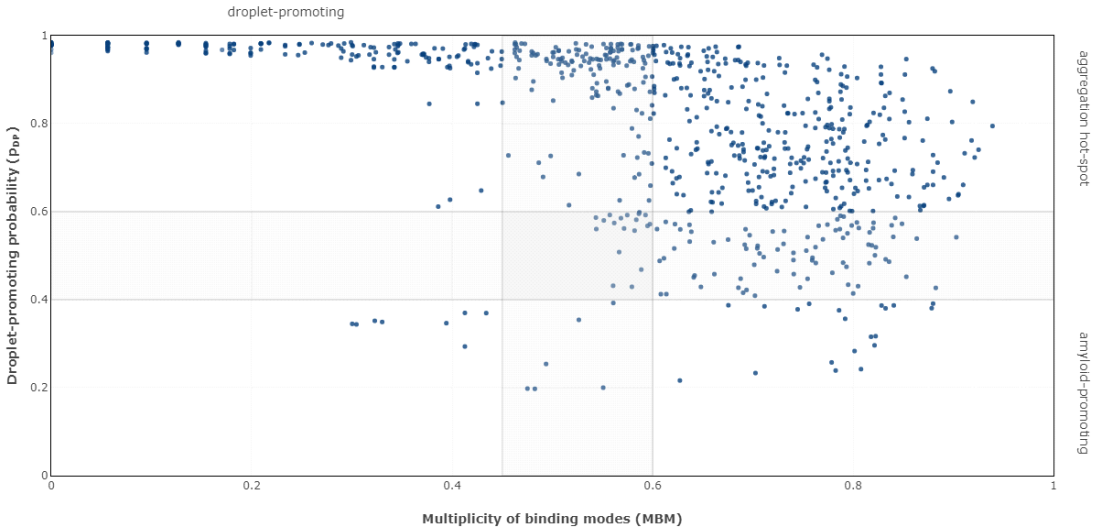
Cyclin DK:


Residue-based droplet-promoting probabilities (p_{DP})
p_{DP}
Droplet-promoting regions (DPRs)
1-27
90-198
Aggregation hot-spots
19-27
93-100
104-111
143-155
167-172
187-198
Multiplicity of binding modes (MBM)
MBM
Regions with context-dependent interactions
19-28
33-52
65-85
93-100
104-111
143-155
167-172
187-198

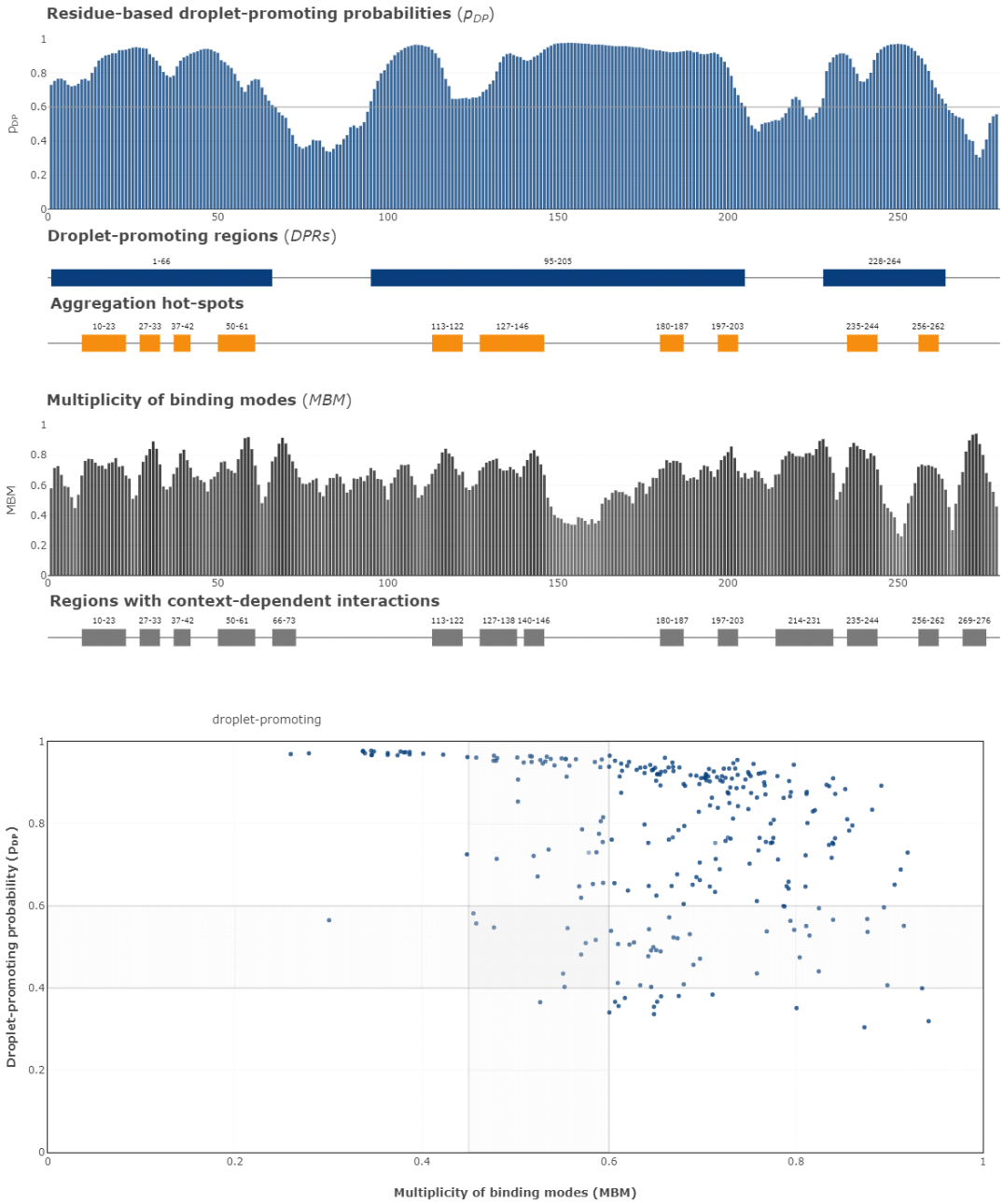

droplet-promoting
aggregation hot-spot
amyloid-promoting
Droplet-promoting probability (p_{DP})
Multiplicity of binding modes (MBM)

AMOTL 2:

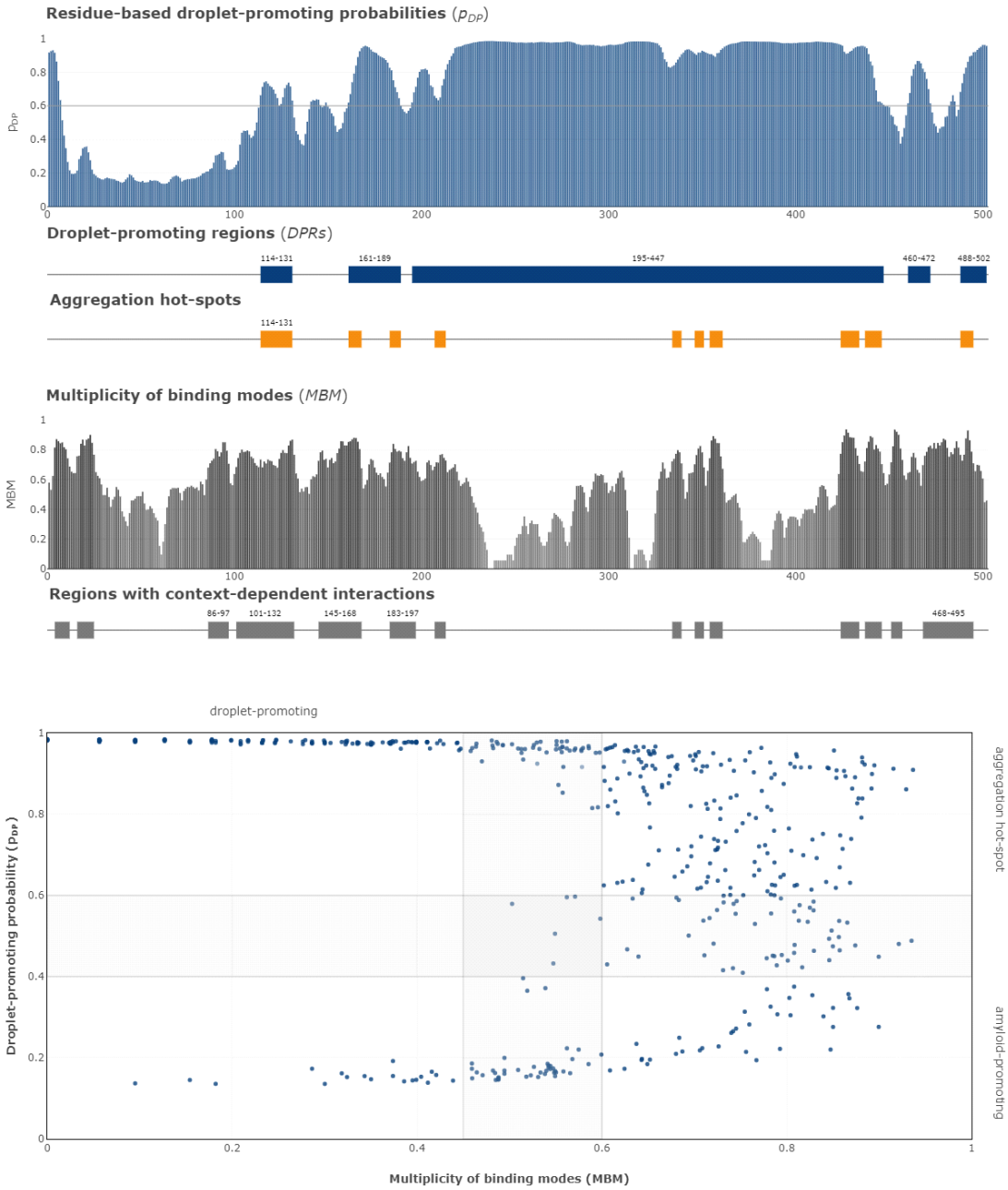

Residue-based droplet-promoting probabilities (p_{DP})
p_{DP}
Droplet-promoting regions (*DPRs*)
31-320
356-417
433-481
502-548
585-629
638-752
Aggregation hot-spots
55-68
235-267
365-399
403-417
470-481
509-523
618-629
Multiplicity of binding modes (*MBM*)
MBM
Regions with context-dependent interactions
6-28
55-68
235-267
365-399
403-421
470-493
509-523
544-566
618-629


droplet-promoting
Droplet-promoting probability (p_{DP})
Multiplicity of binding modes (MBM)
aggregation hot-spot
amyloid-promoting

HCLS 1:


Residue-based droplet-promoting probabilities (p_{DP})
p_{DP}
Droplet-promoting regions (*DPRs*)
1-66
95-205
228-264
Aggregation hot-spots
10-23
27-33
37-42
50-61
113-122
127-146
180-187
197-203
235-244
256-262
Multiplicity of binding modes (*MBM*)
MBM
Regions with context-dependent interactions
10-23
27-33
37-42
50-61
66-73
113-122
127-138
140-146
180-187
197-203
214-231
235-244
256-262
269-276
droplet-promoting
aggregation hot-spot
amyloid-promoting
Droplet-promoting probability (p_{DP})
Multiplicity of binding modes (MBM)

Actin - Binding:

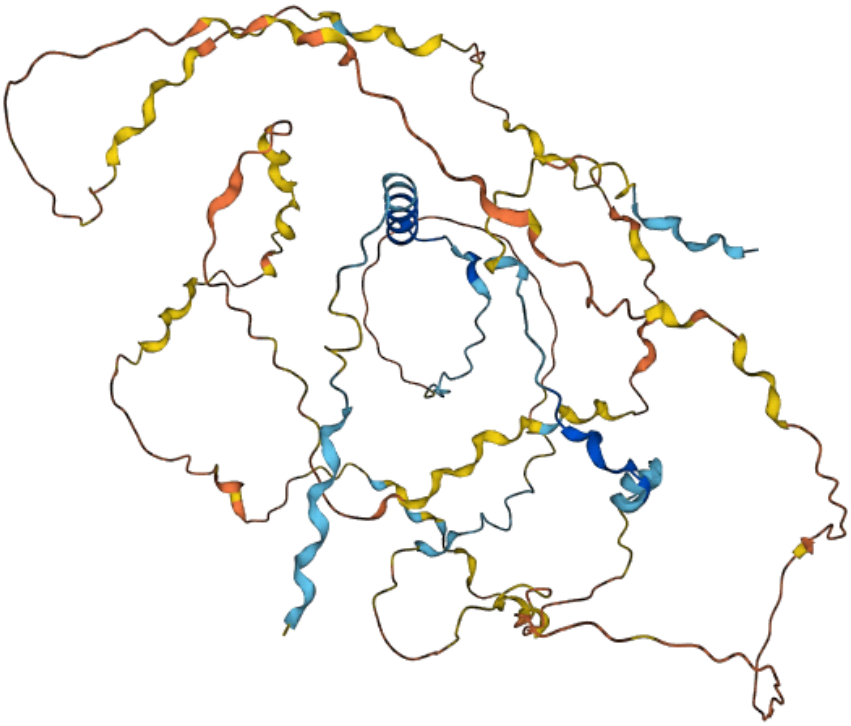


ALPHAFOLD :

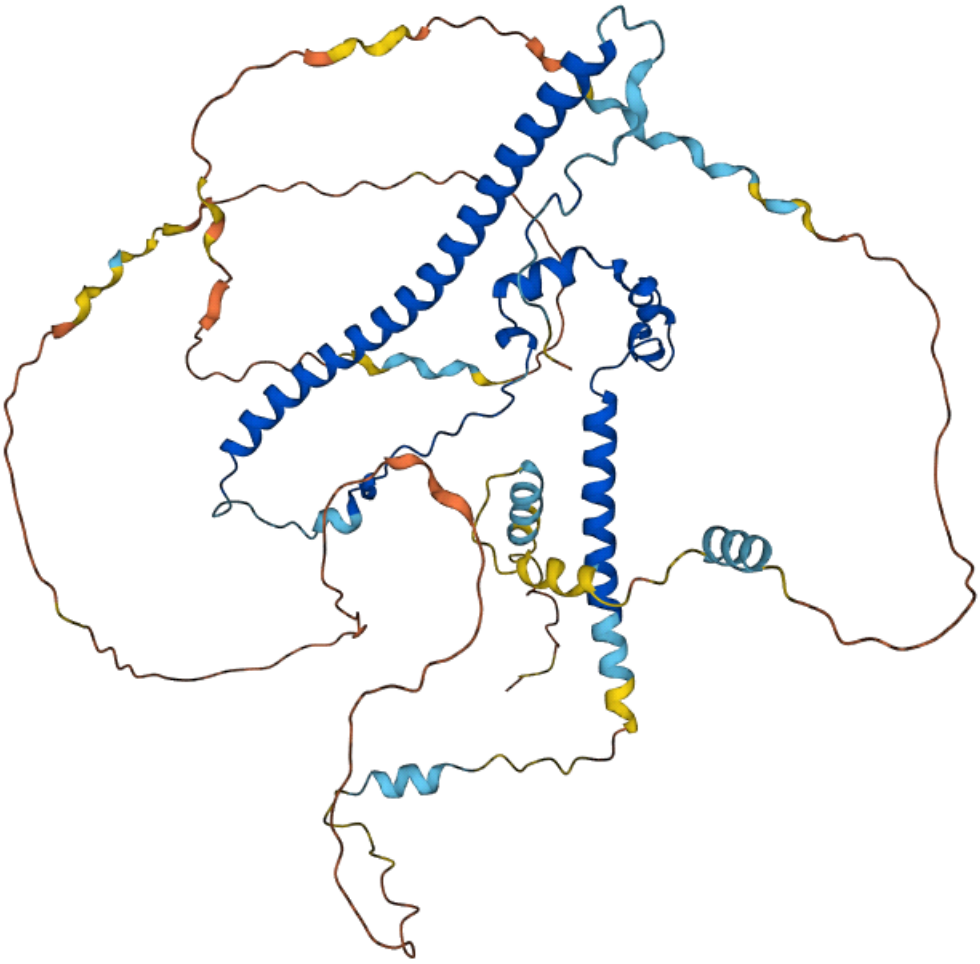
Alphafold is a learning techniques to predict protein structures accurately from their amino acid sequences. The tool employs a sophisticated neural network architecture, trained on a vast dataset of known protein structures, to generate 3D models with remarkable precision. Utilizes a convolutional neural network (CNN) architecture, specifically designed to infer the complex spatial arrangements of atoms within a protein from its sequence. Its performance has been validated in blind assessments like the Critical Assessment of Structure Prediction (CASP) competition.

Following are the Results for Proteins using ALPHAFOLD predictor tool:

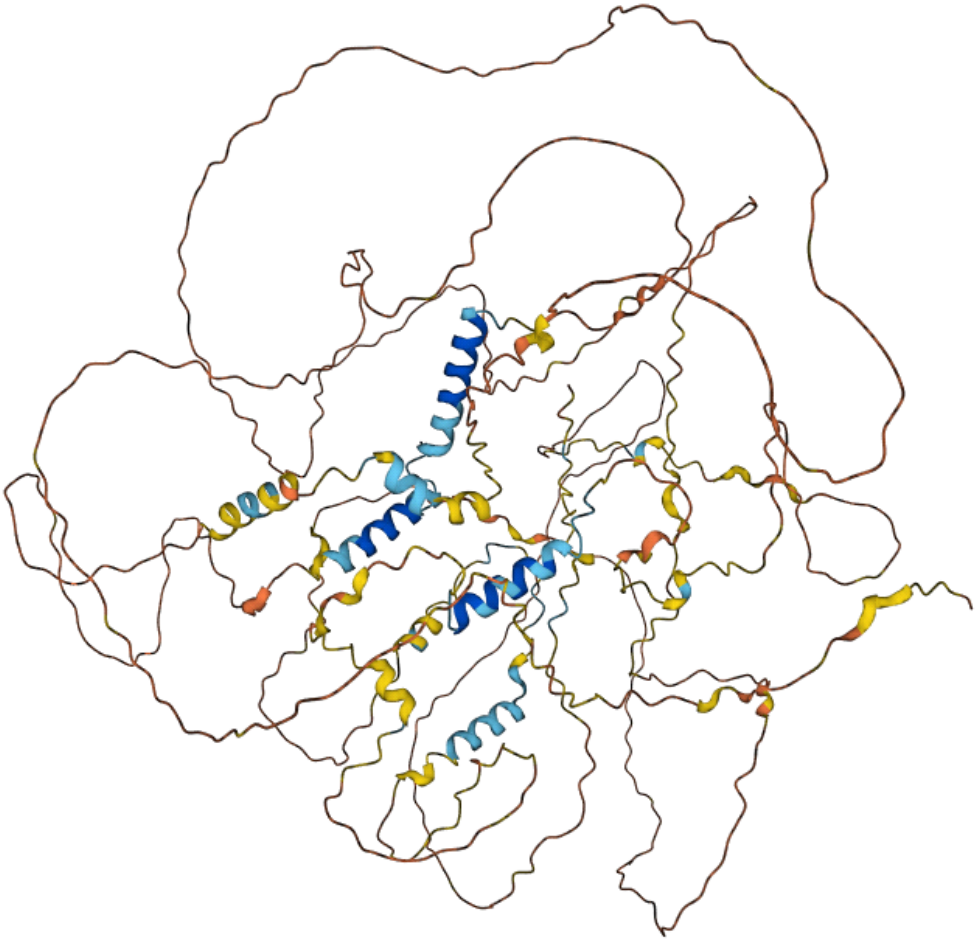
WIPF 1:



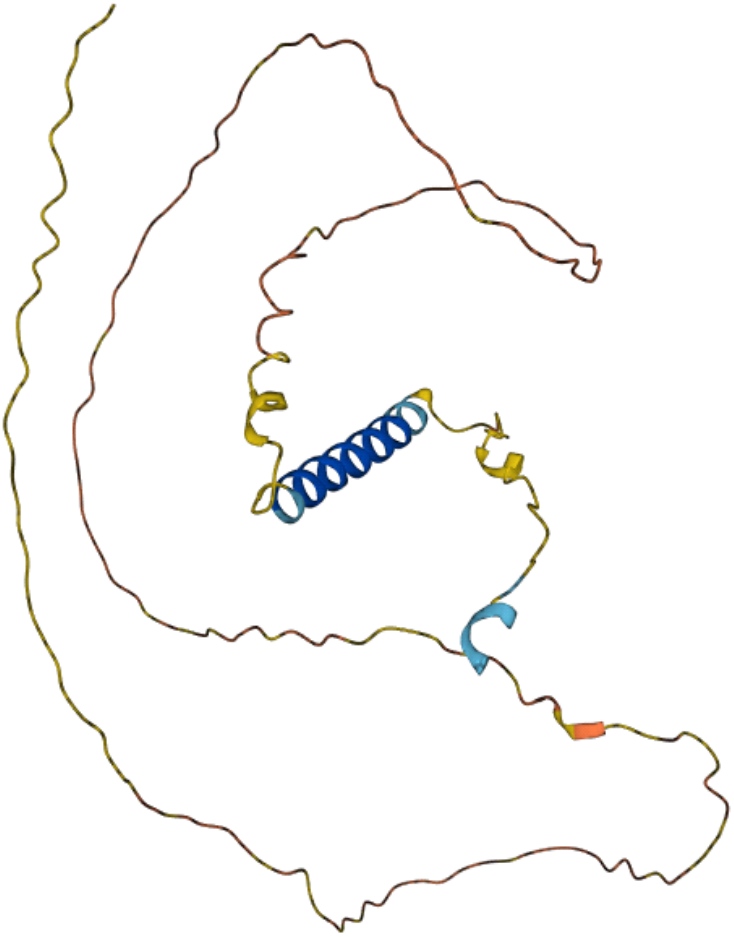
Synaptopodin:



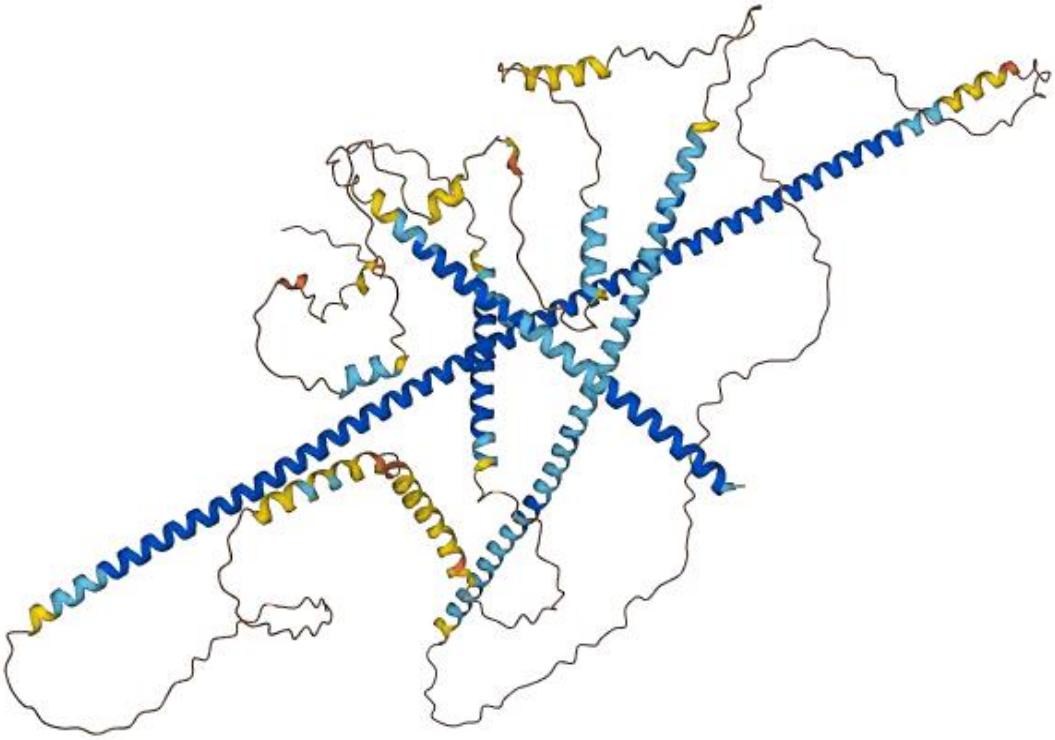
Caldesmon:



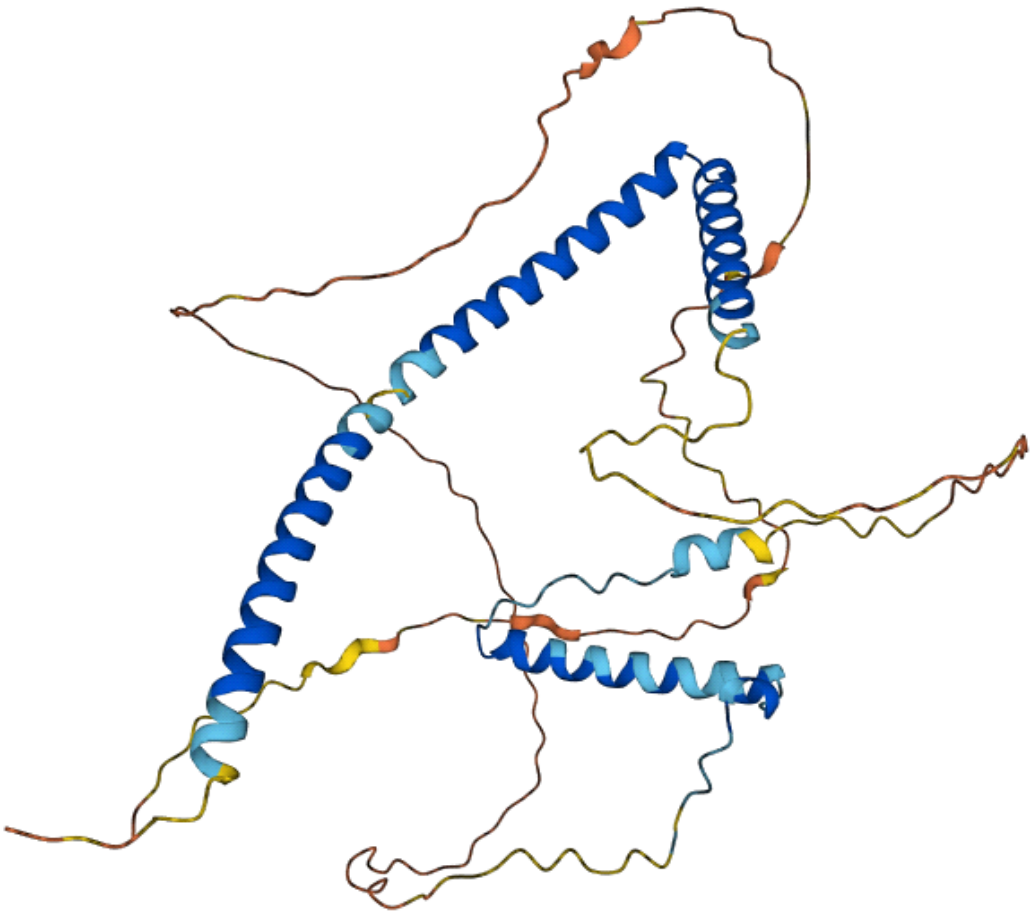
Neuromodulin:



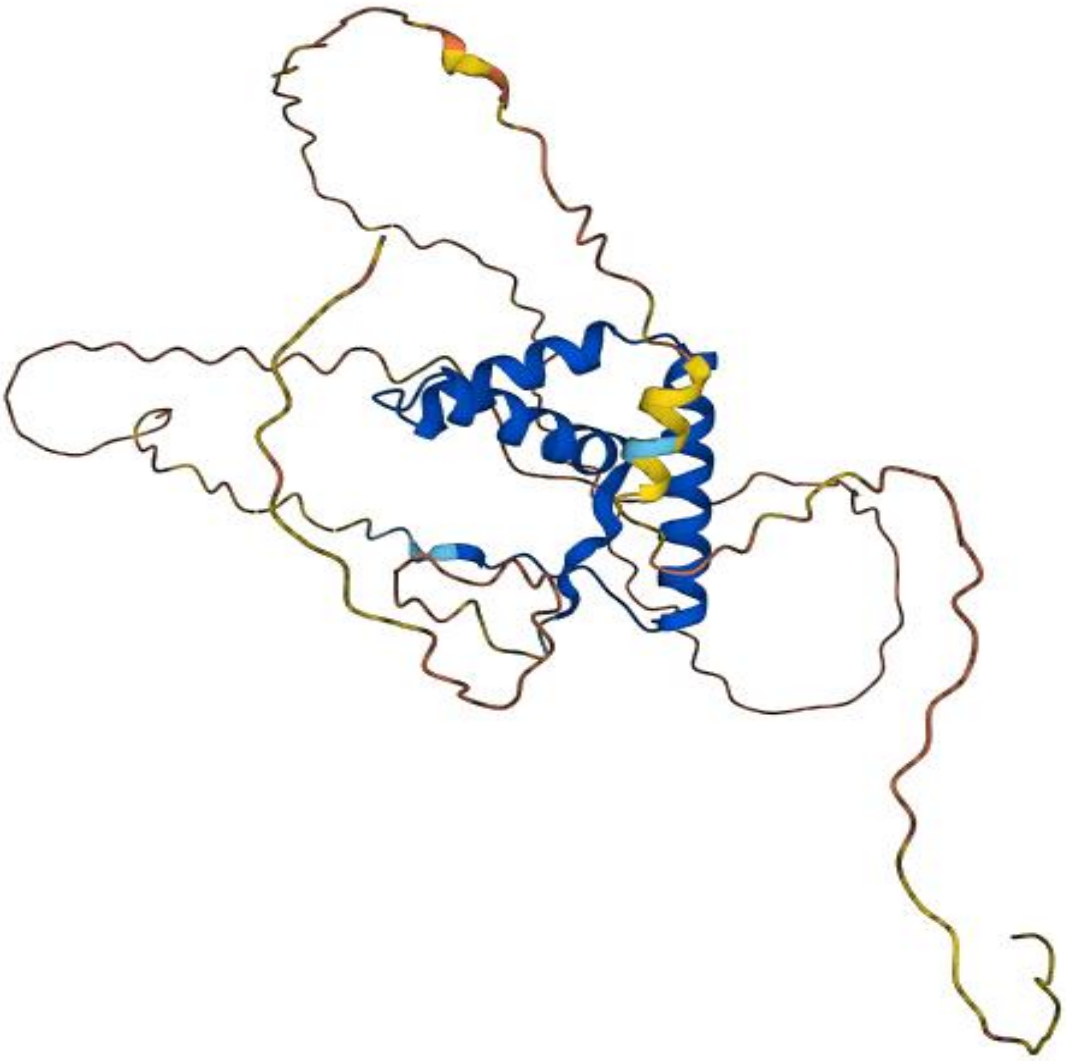
HEXIM 1:



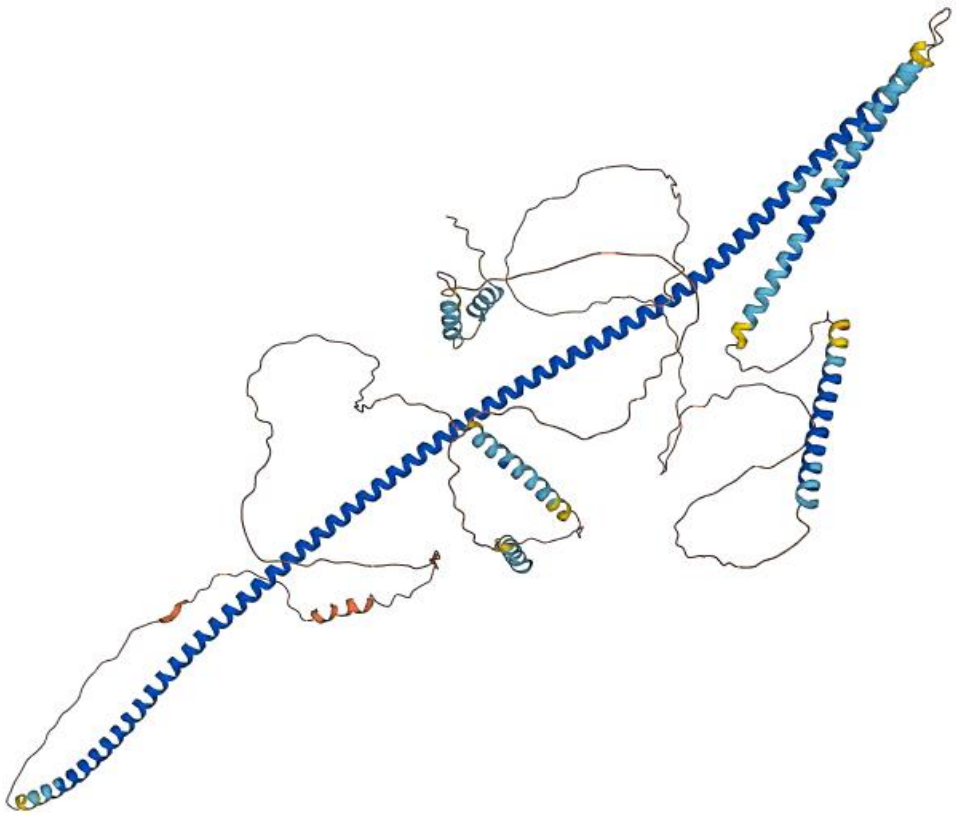
SOX 2:



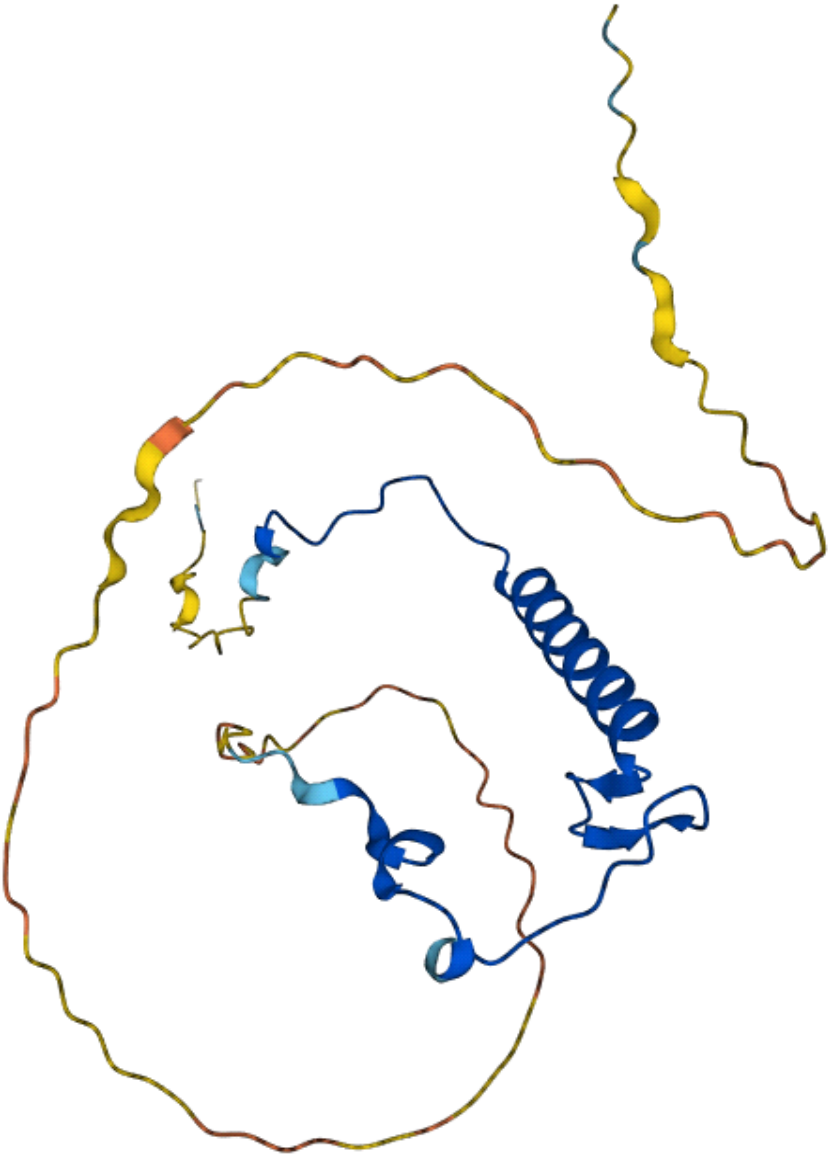
Cyclin DK:



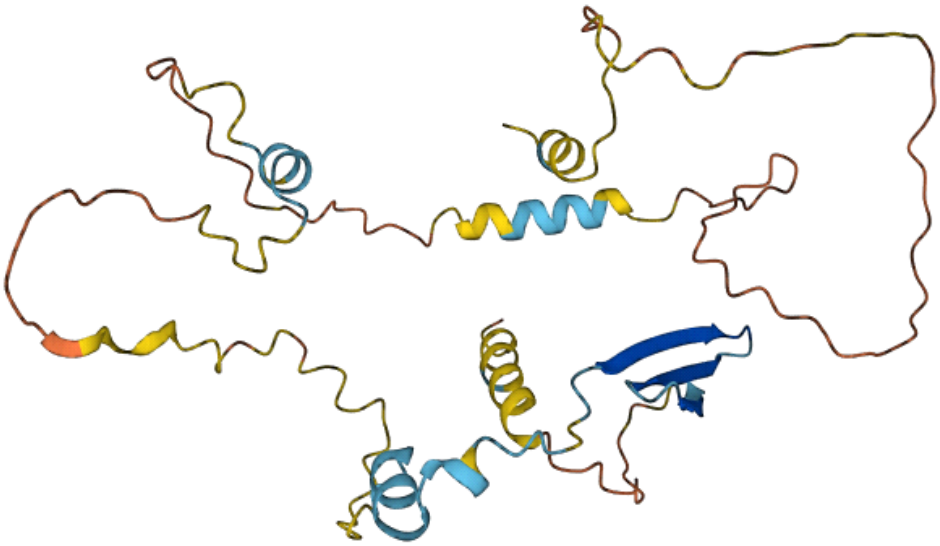
AMOTL 2:



HCLS 1:



Actin - Binding:

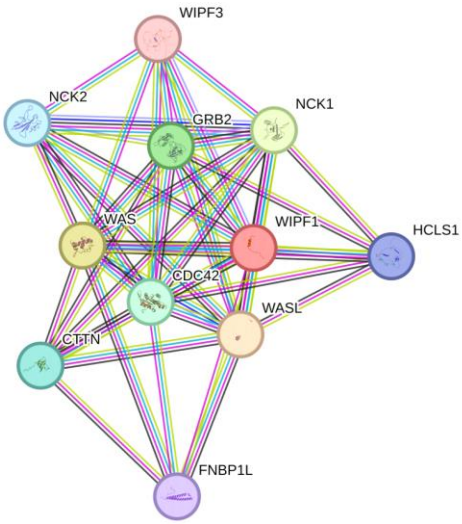


STRING :

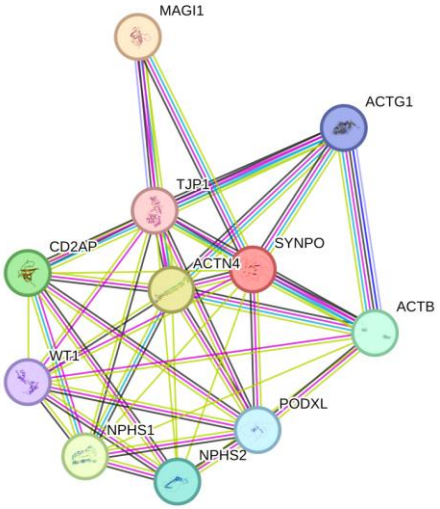
It integrates data from various sources, including experimental studies, computational predictions, and curated databases, to generate networks of interactions between proteins. STRING provides valuable insights into protein functions, pathways, and the organization of cellular processes.

Following are the Results for Proteins using STRING predictor tool:

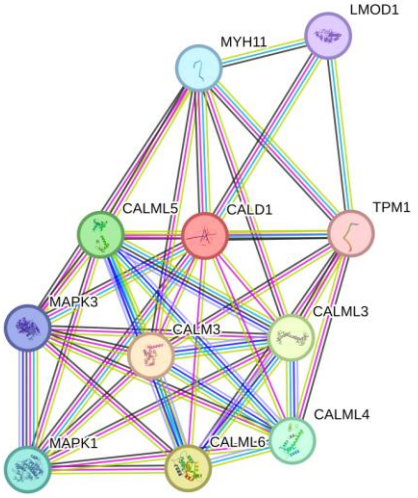
WIPF 1:



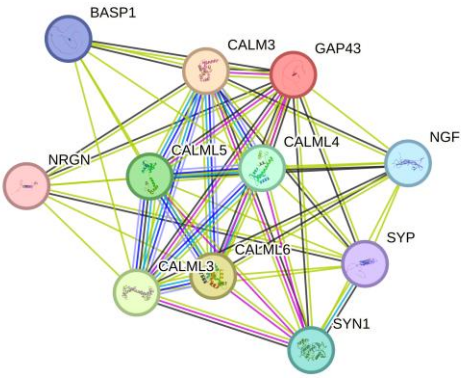
Synaptopodin:



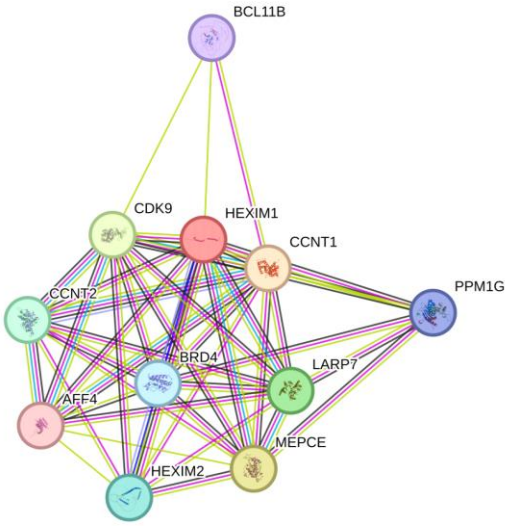
Caldesmon:



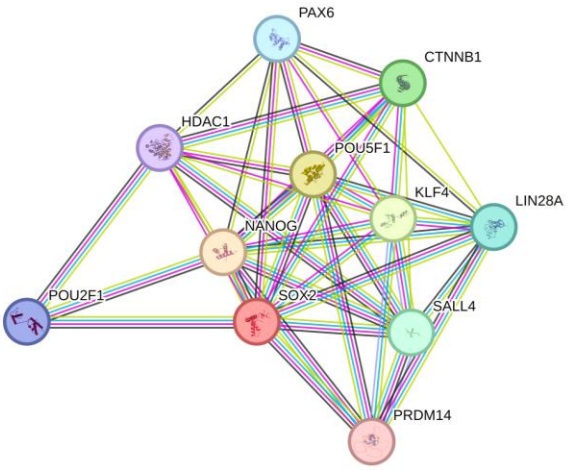
Neuromodulin:



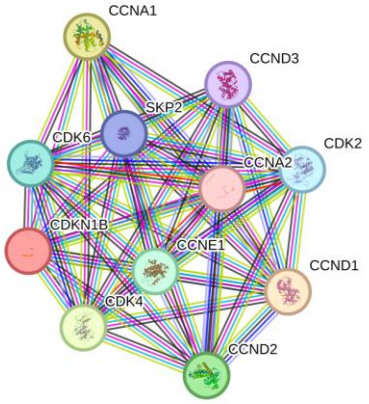
HEXIM 1:


BCL11B
CDK9
HEXIM1
CCNT1
PPM1G
CCNT2
BRD4
LARP7
AFF4
MEPCE
HEXIM2

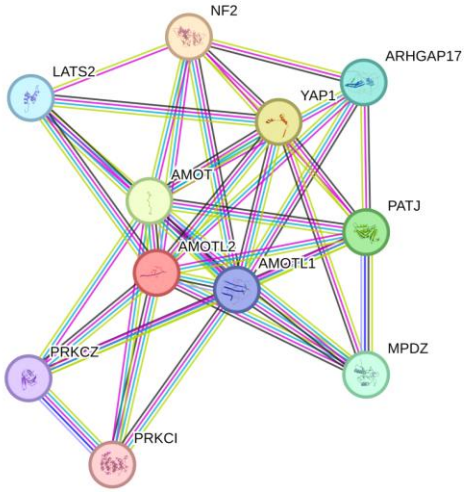
SOX 2:


PAX6
CTNNB1
HDAC1
POU5F1
KLF4
LIN28A
NANOG
POU2F1
SOX2
SALL4
PRDM14

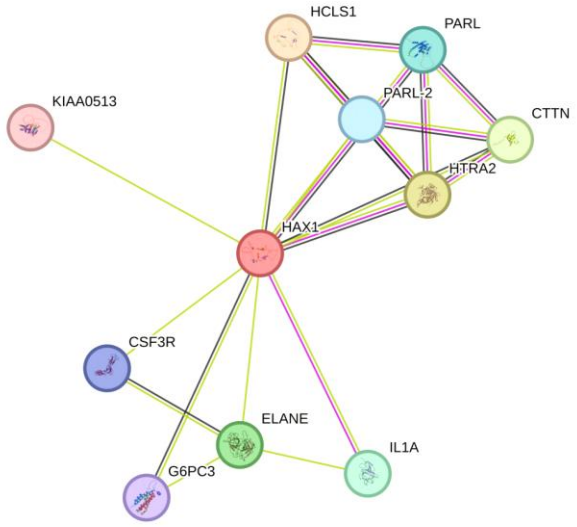
Cyclin DK:



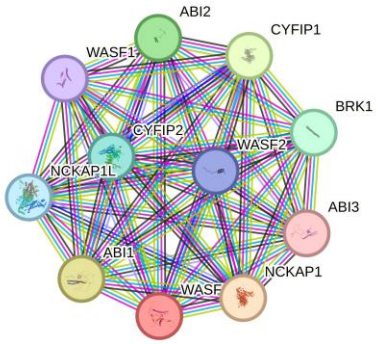
AMOTL 2:



HCLS 1:



Actin - Binding:



DISCUSSION:

I took advantage of RIDAO, D2P2, and FuzDrop as predictor tools. The disorder profiles for each derived protein are shown as graphs in the RIDAO predic on results. It further shows the percentage of disorder in all proteins using a variety of predictors, such as VLXT, VSLB2B, VL3, IUP_S, IUP_L, PFIT, and MDP. On the other hand, D2P2 results show the functional disorder profiles of two proteins using various predictors, including IUPred, Espritz, PV2, PrDOS, VLXT, and VSL2B.

Additionally, it shows where PTMs, MoRFs, and functional domains are posi oned. I have used tools like such as RIDAO, D2P2,FuzDrop, Alphafold and String. The results of RIDAO tool shows the disorder regions for all proteins in the form of Graph. Moreover, it also shows percentage of all proteins using different predictors which includes VLXT, VSLB2B, VL3, IUP_L, PFIT and MDP. Next, D2P2 outcomes show functional disorder pictures for 7 proteins using different predictors such as VLXT, VSL2B, PrDOS, PV2, IUPred and Espritz. Moreover, it also shows the position of functional domains, MoRFs and PTMs. And as last step I used FuzDrop for the probability of proteins for liquid-liquid phase separation.

CONCLUSION AND FUTURE ASPECTS :

Implementing RIDAO, D2P2, Alphafold, String and FuzDrop to perform disorder region analysis in proteins is a significant and comprehensive method of exploring the structural dynamics and functional importance of these distincive proteins. The presence and properties of naturally disordered areas in distinct cortactin proteins can be enhanced by researchers by utilising the capabilities of RIDAO, D2P2, FuzDrop,ALphafold and String. Through the procedure of mapping disorder regions onto the sequence of radiotrophic protein molecules, the intrinsically disordered regions and their interactions are made apparent. This mapping enables simpler to analyse the structural alterations in the protein and how they exhibit spontaneously throughout the organism.

Researching cortactin protein disorders regions will significantly advance our knowledge of cellular control, disease processes, and protein structure-function interactions. The field of biomedical research and therapeutic treatments may be significantly impacted in the future by the application of this data for biotechnological applications, biomarker identification, and the development of pharmaceuticals. It is expected that the research of disorder areas in cortactin proteins will continue to increase in importance and intrigue as science, technology, and computational techniques progress.

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